
LC Electromagnetism I Lecture Notes

Year 1 Semester 2

Ash Stewart

MSci Physics with Particle Physics and Cosmology

School of Physics and Astronomy
University of Birmingham

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Mon 19 Jan 2026 11:00

Lecture 1 - Electrostatics I: EM1 Introduction

1 Course Intro

Course Materials:

- Background material and derivations etc on PowerPoint.
- Worked examples etc are handwritten on visualiser, these are the bits we really need to know.

Why is EM important?

- Foundations of modern technology and the modern world.
- What gives elements their properties.
- Responsible for life itself.
- Everyday materials are held together by EM forces.
- Optics can only be understood through EM theory.

The course aim is to lay down the foundations, eventually leading us to Maxell's Laws.

1.1 Maxwell's Laws

Maxwell's four equations are:

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0} \\ \nabla \wedge \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \cdot \mathbf{B} &= 0 \\ \nabla \wedge \mathbf{B} &= \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)\end{aligned}$$

Where:

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$$

Together, these show that the electric and magnetic fields are related and are two aspects of a single force, the electromagnetic force. We don't have to properly understand them yet, but cannot learn them in EMII unless we fundamentally understand E and B fields from this module.

1.2 Course Structure

Part I: Electric Fields

- Charge and Coulomb's Law.
- The electric field.
- Gauss' Law.
- Capacitors.

Part II: Magnetic Fields

- Magnetic Fields

- Charged Particles in B-Fields
- Electromagnetic Induction.
- Magnetic Dipoles

In this lecture:

- Introduction to EM.
- Electric charge.
- Force between charges.
- The concept of the Electric Field (E-Field).

2 Electric Charge

First attributed to Thales circa. 624 - 546 BC. Experiments by Franklin and Coulomb expanded and showed that there was two types of charge, which they called positive and negative.

The “positive electricity” came from rubbing a glass rod with silk, and the negative from rubbing an ebonite (early plastic) rod with fur. They found that like charges repel and opposite charges attract.

We know that the elementary unit charge is the magnitude of charge of an electron/proton and everything else is a multiple of this ¹:

$$e = 1.6 \times 10^{-19} C$$

This has units of the Coulomb.

2.1 Charge Conservation

Electrons and protons are both stable (protons decay with a life greater than 10^{31} years). This means that the total charge of an isolated system is constant and can be conserved.

They have the same magnitude of charge, exactly:

$$|q_p| = |q_e| = e$$

2.2 Electrostatic Force

Like charges repel and opposite charges attract, along the line of action given by a line drawn between the two charges. The force is proportional to the product of charges so:

$$F \propto q_1 q_2$$

Here, a negative force means attraction and a positive force means repulsion. Newton called this “force at a distance”. Like gravity, two charges will exert a force on each other at a distance without any contact.

There must, therefore, be something between them that mediates this force. Later physics gives this as “virtual particles” which isn’t a Y1 topic, so classically we say that this medium is the Electric Field.

2.3 Electric Field

A charge produces a field around it. Another charge also interacts with this field, and this interaction is what causes a force:

$$\underline{F} = \underline{E}q$$

Where F is the force exerted on a test charge of charge q by a charge Q producing a field E . The magnitude of the electric field has units NC^{-1} (force per units charge).

$$|\underline{E}| \propto Q \quad |\underline{F}| \propto Qq$$

¹While quarks have fractional charge, we don’t get free quarks

Consider a point charge with a spherical electric field spreading out around it. As the distance from the charge increases, the surface area increases as $4\pi r^2$. Therefore the magnitude of the electric field must decrease with $4\pi r^2$

Therefore:

$$|\underline{E}| \propto \frac{Q}{4\pi r^2}$$

We need a (inverse) constant of proportionality. This depends on the medium, but for a vacuum we call it the permittivity of free space ϵ_0 :

$$\epsilon_0 = 8.854 \times 10^{-12} C^2 m^{-2} N^{-1}$$

Hence:

$$E = |\underline{E}| = \frac{Q}{4\pi\epsilon_0 r^2}$$

2.4 Direction of the E-Field

Force is a vector, so the E-field must be too. Consider an E-field from a charge Q at distance r . We give E components E_r , E_θ , E_ϕ , where, since it's a sphere ϕ and θ represent the unit vectors in the two possible tangential directions.

If there was a component in E_θ this would be clockwise from one perspective, but anticlockwise from another (walking behind it). This is not possible, as the field must behave in the same manner from all viewpoints. Therefore:

$$E_\theta = E_\phi = 0$$

$$\underline{E} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r}$$

2.5 Force between two charges

Consider two charges q_1 and q_2 . The force on q_2 due to the e-field from q_1 is:

$$\underline{F}_1 = \underline{E}_1 q_2$$

This is equal to the force on q_1 due to the e-field from q_2 , given by:

$$\underline{F}_2 = \underline{E}_2 q_1$$

So the force between two charges is:

$$\underline{F} = \frac{q_1 q_2}{4\pi\epsilon_0 r^2} \hat{r}_{12}$$

2.6 Force between many charges

If we have more than two positive charges, we use the “principle of superposition”. Effectively, you consider each pair of charges at the same time and vector sum of the forces together. I.e. if we have three points and we care about the net force on one, we take the vector sum of the two vectors from that point to the others.

In general:

$$\begin{aligned} \underline{F} &= \underline{F}_1 + \underline{F}_2 + \underline{F}_3 + \dots \\ &= q \sum_i \frac{q_i}{4\pi\epsilon_0 r_i^2} \hat{r}_i \end{aligned}$$

Where r_j is the distance between q_i and q , with unit vector $\hat{r}_j$ between them.

Since $\underline{F} = q\underline{E}$, the electric field at a test charge q must be:

$$\underline{E} = \sum_i \frac{q_i}{4\pi\epsilon_0 r_i^2} \hat{r}_i$$

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Lecture 2 - Electrostatics II: Field Lines and Continuous Charges

1 Field Lines

To visualise a field, we draw field lines. These are drawn so that the tangent of these lines at any point in space gives the direction of $\underline{E}$ at that point.

The lines point radially outwards from a positive charge, and radially towards a negative charge. The density of these lines is proportional to the magnitude of $\underline{E}$, but the absolute number of lines doesn't convey much.

Field lines must start at a positive charge and end on a negative charge. They are continuous and can never cross each other, they only start/stop at charges or proceed to infinity.

2 Continuous Charge Distributions

What do we do if we have a charge which isn't a point particle?

- We break the object up into a distribution of infinitesimal point charges.
- We treat each piece as a point charge and add up the separate $\underline{E}$ fields across an integral.
- This gets messy for anything non-trivial, resulting in a complex 3D integral often requiring computational techniques.

Charges can be distributed across an object in a number of ways:

- Charge per unit length across a line, given by λ :

$$q = \int \lambda dl$$

- Charge per unit area across a surface area, given by σ :

$$q = \int \sigma dA$$

- Charge per unit volume, given by ρ :

$$q = \int \rho dV$$

3 Example: E-Field from a Line of Charge

Consider an E-field from a line of length $2L$, measured at a point along the midpoint's perpendicular. This point sits x length units along the perpendicular.

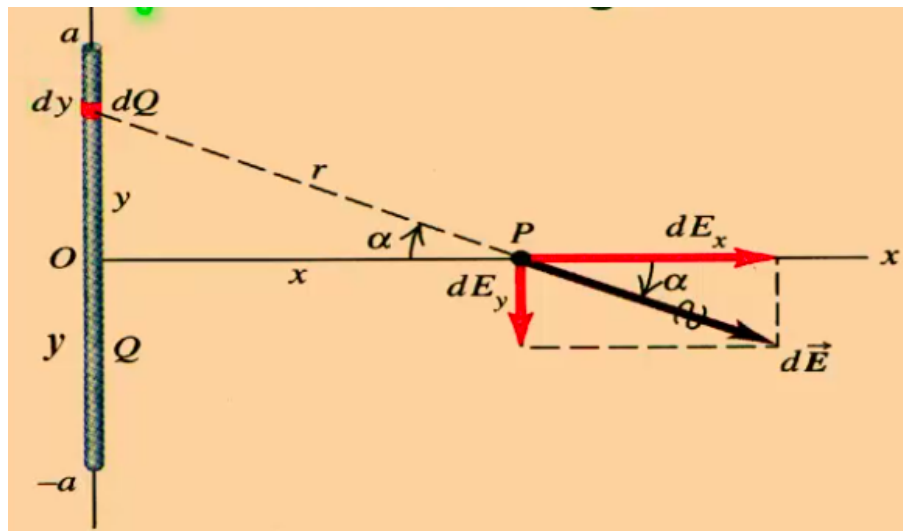


Figure 2.1

We consider some element of charge $\delta Q = \lambda \delta y$ along the line at y . There are two components to the E-field, one pointing in the positive x direction and one in the negative y direction.

When we consider all of the δQ s along the length of the charged rod, each point on the positive y -axis along the length will have a corresponding point on the negative y -axis. These two points will have equal and opposite magnitude, so $E_y = 0$ by symmetry. There is therefore no resultant y component and the final field will point along $\hat{x}$.

For some infinitesimal δq generically along the line of charge:

$$\delta E_x = \frac{\delta q}{4\pi\epsilon_0 r^2} \cos \alpha \quad (1)$$

And for each δQ :

$$\delta Q = \lambda \delta y \quad (2)$$

And from trig:

$$r^2 = y^2 + x^2 \quad (3)$$

$$\cos \alpha = \frac{x}{r} \quad (4)$$

Substituting 2, 3, 4 into 1:

$$\delta E_x = \frac{\lambda \delta y}{4\pi\epsilon_0} \frac{x}{[y^2 + x^2]^{3/2}}$$

Hence:

$$E_x = \frac{\lambda x}{4\pi\epsilon_0} \int_{-L}^L \frac{1}{[y^2 + x^2]^{3/2}} dy$$

To solve this, we would use the standard integral given in the question. Alternatively, if a standard integral isn't given in the question we could express in the integral in terms of α and integrate from the minimum to maximum α .

3.1 Example Part B: The Easier Route

As before:

$$\delta E_x = \frac{\lambda \delta y}{4\pi\epsilon_0 r^2} \cos \alpha \quad (1)$$

$$\cos \alpha = \frac{x}{r} \implies r = \frac{x}{\cos \alpha} \quad (2)$$

And:

$$\tan \alpha = \frac{y}{x} \implies y = x \tan \alpha \quad (3)$$

As:

$$\frac{d}{d\alpha} \tan \alpha = \frac{1}{\cos^2 \alpha}$$

We have:

$$\delta y = \frac{x \delta \alpha}{\cos^2 \alpha} \quad (4)$$

Hence:

$$\begin{aligned} \delta E_x &= \frac{\lambda}{4\pi\epsilon_0} \underbrace{\frac{x \delta x}{\cos^2 \alpha}}_{\delta y} \underbrace{\frac{\cos^2 \alpha}{x^2}}_{1/r^2} \cos \alpha \\ &= \frac{\lambda}{4\pi\epsilon_0 x} \cos \alpha \end{aligned}$$

Which is much easier to integrate from the minimum to maximum values of α .

$$E_x = \frac{\lambda}{4\pi\epsilon_0 x} \int_{\alpha_{\min}}^{\alpha_{\max}} \cos \alpha \, d\alpha$$

And from the geometry of the problem:

$$\alpha_{\max} = \arctan\left(\frac{L}{x}\right) \quad \alpha_{\min} = -\arctan\left(\frac{L}{x}\right)$$

Which is easy to integrate.

If the problem was setup such that the rod was infinitely long, then:

$$\begin{aligned} E_x &= \frac{\lambda}{4\pi\epsilon_0 x} \int_{-\pi/2}^{\pi/2} \cos \alpha \, d\alpha \\ E_x &= \frac{\lambda}{2\pi\epsilon_0 x} \end{aligned}$$

This is approximately true for the case where $x \ll L$.

4 Example: Ring of Charge

Consider a ring of charge with radius a . We step out some perpendicular constant distance x from the centre of the ring to point P . What is the resultant field at P ?

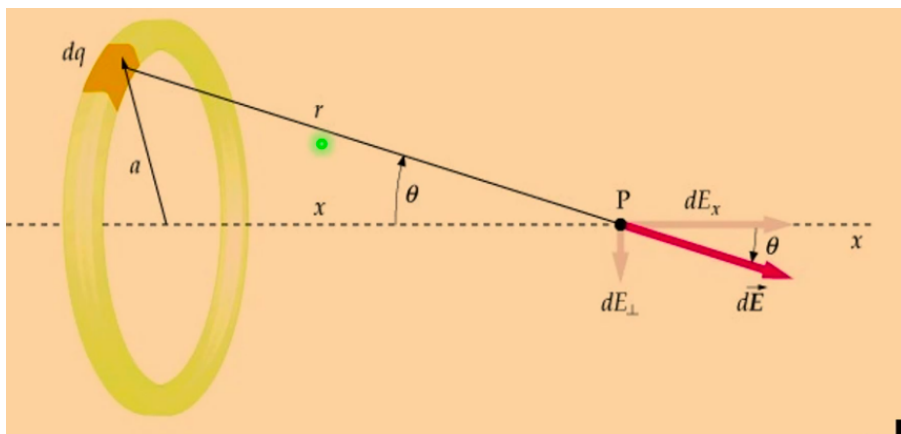


Figure 2.2

The ring has total charge Q , distributed with charge per unit length λ .

Again, we can use symmetry. For every point on the ring, there will be a point one diameter away creating an equal and opposite δE . Hence, the y and z components cancel leaving only E_x , pointing in the $\hat{x}$ direction.

$$E_y = E_z = 0 \text{ by symmetry}$$

We have:

$$\begin{aligned}\delta E_x &= \frac{\delta q}{4\pi\epsilon_0 r^2} \cos \theta \\ \cos \theta &= \frac{x}{r} \\ r &= \sqrt{a^2 + x^2}\end{aligned}$$

Hence:

$$\delta E_x = \frac{\delta q}{4\pi\epsilon_0} \frac{x}{[a^2 + x^2]^{3/2}}$$

Here, a and x are constants, so the only variable we can integrate across is δq :

$$\begin{aligned}E_x &= \frac{x}{4\pi\epsilon_0} \frac{1}{[a^2 + x^2]^{3/2}} \int dq \\ &= \frac{xQ}{4\pi\epsilon_0 [a^2 + x^2]^{3/2}}\end{aligned}$$

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Lecture 3 - Electrostatics III: More Continuous Charges and Gauss' Law

1 Example: Large Circular Plane

Consider a large circular plane with uniform charge density σ . We step out some distance x along the axis perpendicular to the plane, through its centre, to point P .

The plane has radius R and total charge Q . What is the magnitude and direction of the electric field at P ?

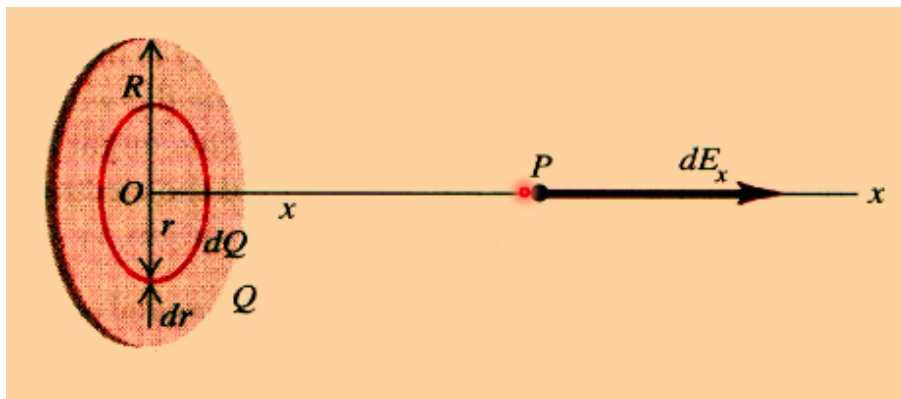


Figure 3.1

Using the same logic as for the ring last lecture, we can see that:

$$E_y = E_z = 0 \text{ by symmetry}$$

So the final E-field lies entirely in the direction of $\hat{x}$.

We want to divide the final E-field into an infinite number of thin rings, each at radius r with infinitesimal thickness δr . We can then integrate across all rings to determine the total E-field.

The distance between P and O is x , the distance between O and the charge element is r , and the distance between P and the charge element is d .

From the previous example (in an exam question we would have to do this too), we have:

$$\delta E_{\text{ring}} = \frac{Q_{\text{ring}}}{4\pi\epsilon_0} \frac{x}{[r^2 + x^2]^{(3/2)}}$$

$$Q_{\text{ring}} = \sigma \times 2\pi r \delta r$$

Hence:

$$\delta E_{\text{ring}} = \frac{\sigma r \delta r}{2\epsilon_0} \frac{x}{(r^2 + x^2)^{(3/2)}}$$

Noting that $r = x \tan \theta$, where θ is the angle between d and x :

$$\delta r = \frac{x}{\cos^2 \theta} d\theta$$

$$\Rightarrow \cos \theta = \frac{x}{d} = \frac{x}{[r^2 + x^2]^{1/2}}$$

$$\frac{1}{[r^2 + x^2]^{3/2}} = \frac{\cos^3 \theta}{x^3}$$

Substituting these in:

$$\delta E_{\text{ring}} = \frac{\sigma}{2\epsilon_0} \underbrace{\left(\frac{x \sin \theta}{\cos \theta} \right)}_r \underbrace{\frac{x \delta \theta}{\cos^2 \theta}}_{\delta r} \times x \times \frac{\cos^3 \theta}{x^3}$$

Which simplifies down to:

$$\frac{\sigma}{2\epsilon_0} \sin \theta \delta \theta$$

Which is again nice and simple to integrate. Integrating across all θ :

$$E_{\text{disc}} = \frac{\sigma}{2\epsilon_0} \int_0^{\theta_{\text{max}}} \sin \theta d\theta$$

Where $\theta_{\text{max}} = \arctan(R/x)$. If we wanted a full infinite plane, we could get this as $R \rightarrow \infty$.

$$\theta_{\text{max}} = \pi/2$$

$$E = \frac{\sigma}{2\epsilon_0} \int_0^{\pi/2} \sin \theta d\theta$$

$$\Rightarrow E = \frac{\sigma}{2\epsilon_0}$$

Similar to the ring, this is approximately true for a finite plane where $x \ll R$. This is interesting because it implies that (for close distances, where $x \ll R$ still holds), the electric field is constant with respect to distance.

2 E-Field in a Hollow Sphere

Consider a hollow sphere with uniform charged shell density σ . Consider an arbitrary point P inside this spherical shell.

We can use solid angles here, which are effectively three dimensional angles. This is still dimensionless, but is given by a subtended area divided by the radius squared:

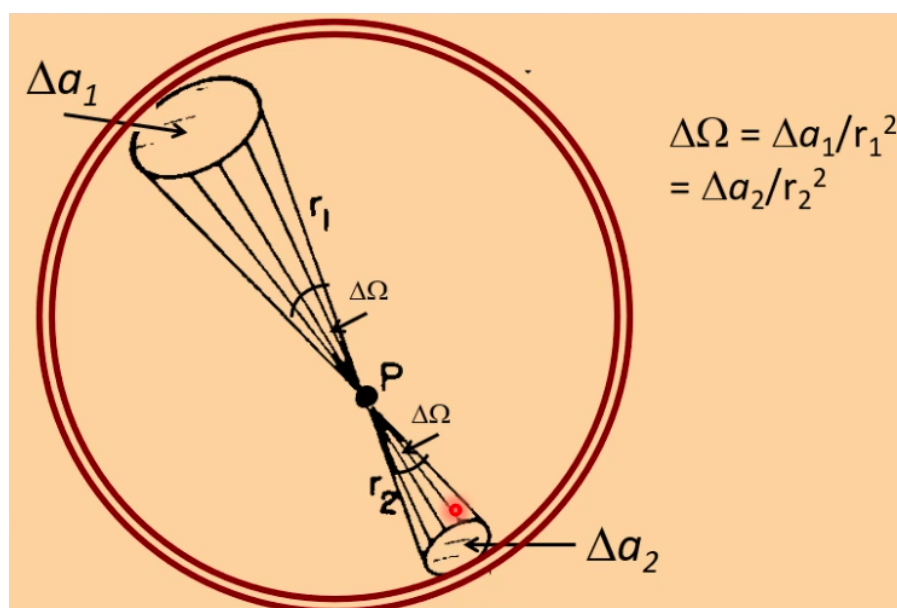


Figure 3.2

While a traditional angle is $\theta = \text{arc length} / \text{radius}$ in units of radians, the A/r^2 for solid angles has units of Steradians, Sr.

The total surface area of a sphere is $4\pi r^2$, so the total solid angle in all direction is 4π .

As we have defined the two cones to subtend the same solid angle at P , their areas scale as the square of their radii:

$$\frac{\Delta a_2}{\Delta a_1} = \frac{r_2^2}{r_1^2}$$

Hence:

$$\frac{\Delta q_2}{\Delta q_1} = \frac{\sigma \Delta a_2}{\sigma \Delta a_1} = \frac{r_2^2}{r_1^2}$$

The ratio of the two electric fields from each area is given by:

$$\frac{E_2}{E_1} = \frac{\Delta q_2 / r_2^2}{\Delta q_1 / r_1^2} = \frac{\Delta q_2}{\Delta q_1} \frac{r_1^2}{r_2^2} = 1$$

This means that the total E-field at P , and indeed at any point inside the hollow sphere, is zero.

3 Electric Flux

Electric flux is effectively the amount of E-field passing across a surface. It is defined in terms of a normal unit vector perpendicular to the surface, $\hat{n}$.

We get an expression for flux by multiplying the field strength by the area "seen" by the field. This is given by $A \cos \theta$ and for a constant E-field on a flat surface:

$$\text{flux} = \Phi_E = \underline{AE} \cdot \hat{n} = EA \cos \theta$$

More generally:

$$\Phi_E = \int \underline{E} \cdot \hat{n} dS = \int \underline{E} \cdot d\underline{S}$$

Where dS is an element of surface area.

4 Gauss' Law

Consider an imaginary closed surface, of any shape. Inside this surface contains a charge q .

The E-field at some element of this surface at distance r from the charge is given by:

$$\underline{E} = \frac{q}{4\pi\epsilon_0 r^2} \hat{r}$$

The flux out of this area element is:

$$d\Phi = \underline{E} \cdot d\underline{S} = \underline{E} \cdot \hat{n} dS = E \cos \theta dS$$

Hence:

$$d\Phi = \frac{q}{4\pi\epsilon_0 r^2} \cos \theta dS = \frac{q}{4\pi\epsilon_0} \frac{\cos \theta dS}{r^2} = \frac{q}{4\pi\epsilon_0} d\Omega$$

Where Ω is the solid angle.

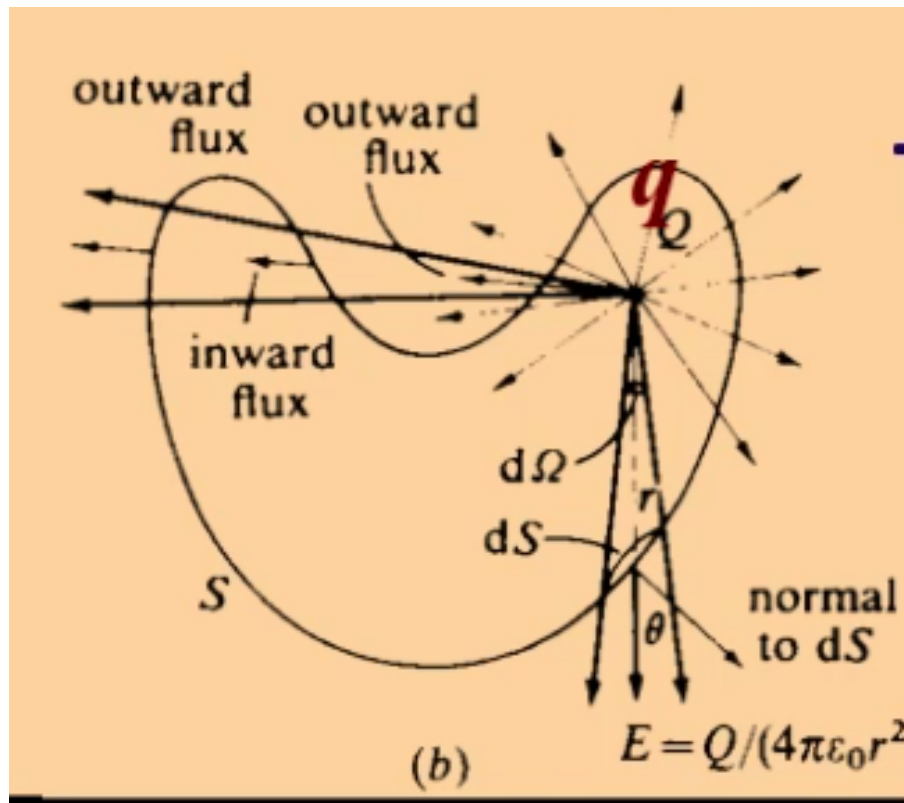


Figure 3.3

The total flux is therefore:

$$\Phi_E = \frac{q}{4\pi\epsilon_0} \int_S d\Omega = \frac{q}{4\pi\epsilon_0} 4\pi = \frac{q}{\epsilon_0}$$

Note that the proof/derivation is not needed, as solid angles are off-spec. Only the final result is needed.

This gives us Gauss' Law:

The net electric flux of $\underline{E}$ out of *any* closed surface, enclosing a total charge of Q_{enclosed} is:

$$\Phi_E = \int_S \underline{E} \cdot d\underline{S} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

Where $Q_{\text{enclosed}} = \sum_i q_i$ for discrete charges or $Q_{\text{enclosed}} = \int_V \rho dV$ for distributed charge.

This becomes incredibly useful for problems where there is symmetry.

4.1 Example: Solid Sphere of Uniform Charge Density

Consider a sphere of radius R with charge Q distributed uniformly. We want to find the E-field inside the sphere, and the E-field outside the sphere.

We will set up an imaginary spherical Gauss surface of radius r and use symmetry:

$$\int_S \underline{E} \cdot d\underline{S} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

Outside the Sphere: $r > R$

The E-field from a sphere and the normal vector of the surface are both radial. Hence, they are parallel. This means that $\theta = 0$ and:

$$\underline{E} \cdot d\underline{S} = E dS$$

Since the sphere is symmetrical, the E-field at a constant r must be constant anywhere around the sphere. Hence, E is constant and can be pulled out of the integral:

$$E \int_S dS = \frac{q_{\text{enclosed}}}{\epsilon_0}$$

$$E(4\pi r^2) = \frac{q_{\text{enclosed}}}{\epsilon_0}$$

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$

Which is Coulomb's law! Lets now consider the E-field inside the sphere.

Inside the Sphere: $r < R$

$$\int_S \underline{E} \cdot d\underline{S} = \frac{q_{\text{enclosed}}}{\epsilon_0}$$

As before, the LHS is $E4\pi r^2$. The RHS is now different, as a smaller volume of charge is enclosed by the Gauss surface. This new charge is given by the total charge times the ratio of the Gauss surface volume to the total volume:

$$\frac{q_{\text{enclosed}}}{\epsilon_0} = \frac{1}{\epsilon_0} \left(\frac{Qr^3}{R^3} \right)$$

Hence:

$$E4\pi r^2 = \frac{Qr^3}{\epsilon_0 R^3}$$

$$\implies E = \frac{Qr}{4\pi\epsilon_0 R^3}$$

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Lecture 4 - Electrostatics IV: Gauss' Law Examples

1 Solid Sphere with Non-Uniform Charge

What if we have a solid sphere with distributed charge where the charge distribution is a function of radius?

$$\rho(r) = \rho_0 r$$

Where r_0 is a constant (note: does not have dimensions of charge density).

We now have:

$$Q_{\text{enclosed}} = \int_0^r \rho(r) dV$$

Hence:

$$\int_S \underline{E} \cdot d\underline{S} = \frac{1}{\epsilon_0} \int_0^r \rho(r) dV$$

We divide the sphere into thin shells of infinitesimal thickness δr . The volume of each shell is:

$$\delta V = 4\pi r^2 \delta r$$

Hence:

$$\begin{aligned} Q_{\text{enclosed}} &= \int_0^r \rho(r) dV = \int_0^r \rho_0 r 4\pi r^2 dr \\ &= 4\pi \rho_0 \int_0^r r^3 dr = \pi \rho_0 r^4 \end{aligned}$$

Using Gauss' law:

$$\int_S \underline{E} \cdot d\underline{S} = \frac{\pi \rho_0 r^4}{\epsilon_0}$$

Considering the LHS and using symmetry:

$$\int_S \underline{E} \cdot d\underline{S} = \int_S E dS = E \int_S dS = E 4\pi r^2$$

Hence:

$$\begin{aligned} 4\pi E r^2 &= \frac{\pi \rho_0 r^4}{\epsilon_0} \\ \implies E &= \frac{\rho_0 r^2}{4\epsilon_0} \end{aligned}$$

2 E-Field Inside Uniform Charged Shell

We've already done this with solid angles, but it becomes much easier with Gauss' law. Since the Gauss surface sits inside the shell, none of the charged distributed across the shell is inside the surface. Therefore, $Q_{\text{enclosed}} = 0 \implies E = 0$

3 E-Fields in Conductors

A simple model of a conductor is where each atom has a free valance electron which is free to move throughout the conductor, in vast quantities.

If you place a conductive box inside a conductor, the electrons want to travel against the field. This leaves a positively charged nuclei, which drift in the opposite direction.

This causes the outside of the metal box to pick up a small charge, creating its own electric field which opposes the initial field. This continues until it perfectly cancels out the initial field, leaving no resultant field inside a conductor.

This is called a “Faraday Cage” and can be used to shield the interior of the conducting box from the E-field, i.e. to protect sensitive electronics.

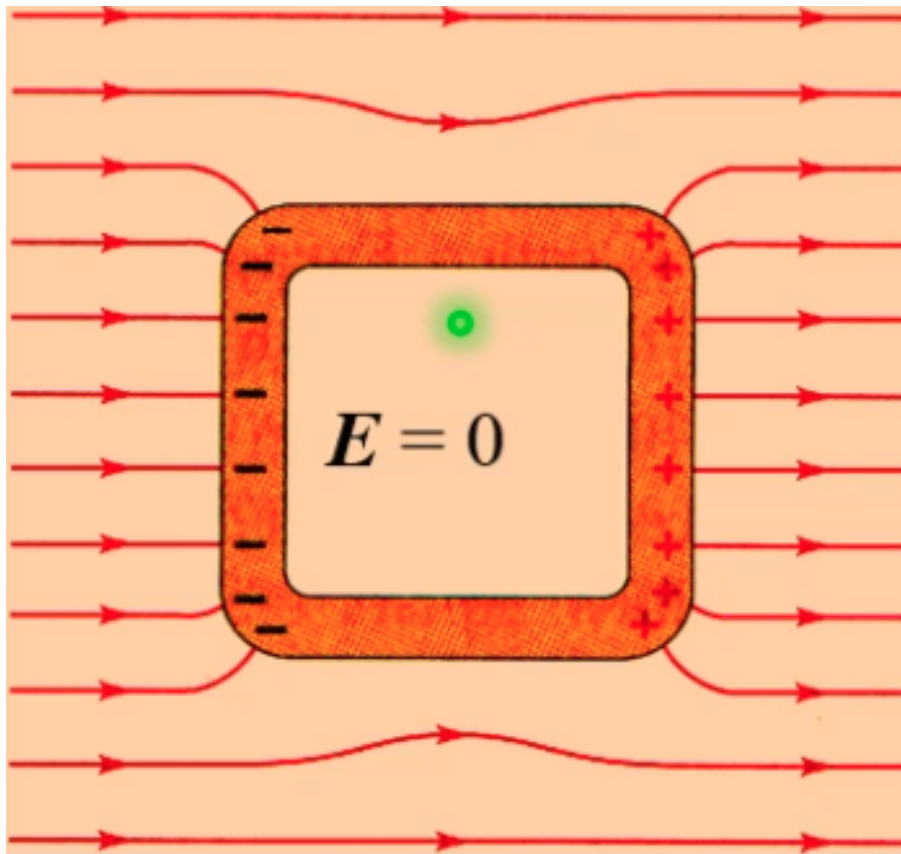


Figure 4.1

4 Infinite Charged Sheet

4.1 Non-Conducting Sheet

Consider an infinitely large plane of surface charge density σ . As the sheet is infinitely large, there's no edge effects and the E-field exists exclusively $\perp$ to the surface of the sheet.

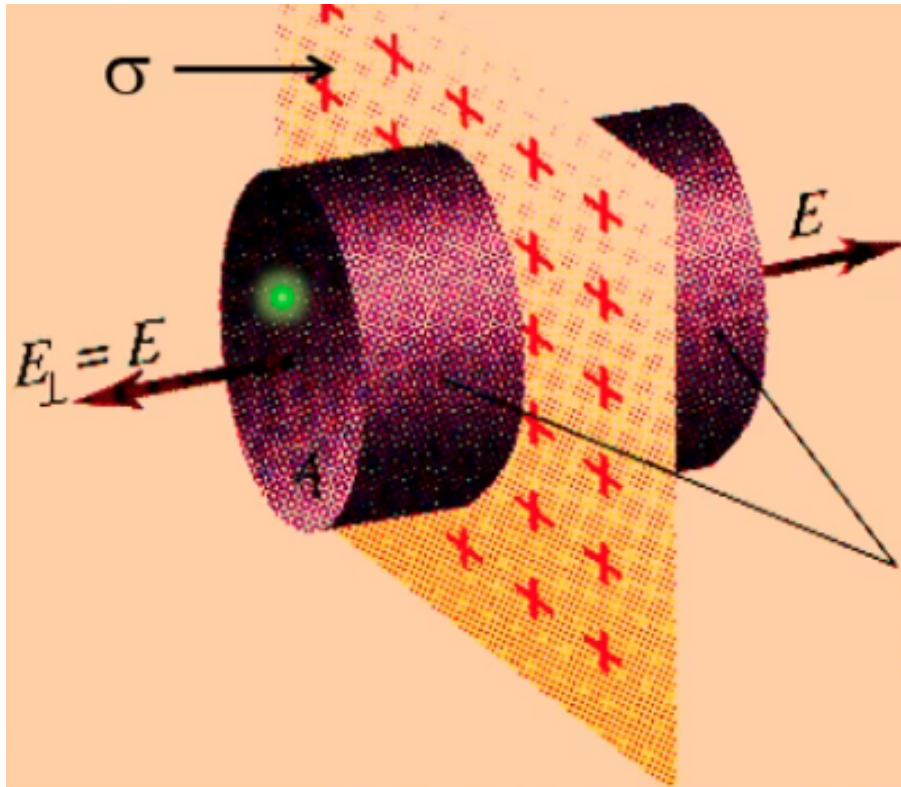


Figure 4.2

We can therefore take a cylindrical Gauss surface, and the only E-field will exist normally to the end circular faces (of cross-sectional area A). Hence, $\underline{E}$ and $d\underline{S}$ are parallel, and:

$$\int_S \underline{E} \cdot d\underline{S} = E \int_S dS$$

So:

$$\begin{aligned} E \int_S dS &= \frac{Q_{\text{enclosed}}}{\epsilon_0} \\ E(2A) &= \frac{\sigma A}{\epsilon_0} \\ \Rightarrow E &= \frac{\sigma}{2\epsilon_0} \end{aligned}$$

As previously derived! Again, this works for a finite plane where the plane is much larger (relatively) than the distance from the plane, so edge effects can be disregarded.

4.2 Conducting Sheet

As the sheet is now conducting, there is no E-field inside the conductor, and charge is distributed to build up a charge density (σ) on both surfaces of the sheet.

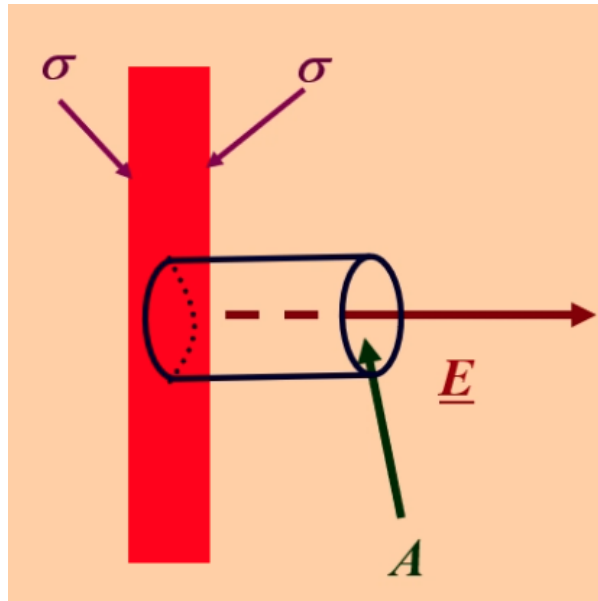


Figure 4.3

Applying Gauss' law gives:

$$EA = \frac{\sigma A}{\epsilon_0}$$

$$\Rightarrow E = \frac{\sigma}{\epsilon_0}$$

Which is a factor of two different from the non-conducting case. Note that this doesn't arise because we only took a Gauss surface on one side. If we took a Gauss surface through the sheet (in the same manner as the non-conducting case), we would have two sets of charge distributed, one on each side, both with charge density σ . We would therefore have:

$$E(2A) = \frac{2\sigma A}{\epsilon_0}$$

Which yields the same result.

5 E-Field Between Charged Conducting Plates

Consider a capacitor-like setup with two parallel charged metal plates.

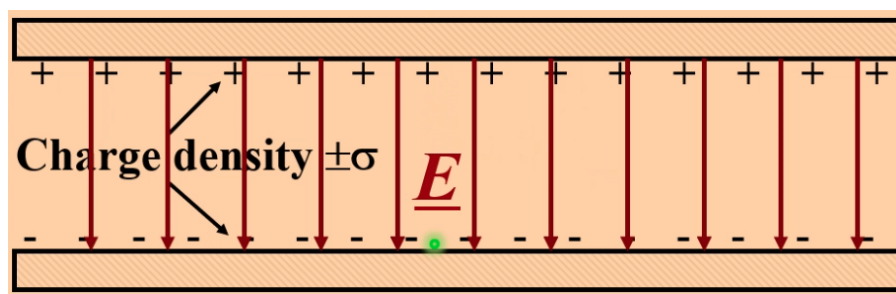


Figure 4.4

While a full derivation will be covered later, the E-field given by this is the same as for an infinite sheet:

$$E = \frac{\sigma}{\epsilon_0}$$

6 Infinite Line of Charge

Consider an infinitely long line charge, of charge density λ . As its infinitely long, there's no edge effects and the electric field must be radial.

We can place a cylindrical Gaussian surface where the axis is parallel to the line, of length L and radius r .

Since the field is radial, it acts in the same direction as the normal vector to the cylinder surface. By symmetry, the E-field is constant for fixed radius, so:

$$\int_S \underline{E} \cdot d\underline{S} = E \int_S dS = E(2\pi rL)$$

$$\frac{Q_{\text{enclosed}}}{\epsilon_0} = \frac{\lambda L}{\epsilon_0}$$

Hence:

$$E(2\pi rL) = \frac{\lambda L}{\epsilon_0}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

7 No Contained Charge

What if the Gaussian surface sits within an E-field, but the charged creating the field is not enclosed in the surface?

This could be thought of as implying by Gauss' law that:

$$\oint \underline{E} \cdot d\underline{S} = 0$$

$$\implies E = 0?$$

But this is not the case. Here, the symmetry that we've relied upon isn't here, as $\underline{E} \nparallel d\underline{S}$ and is not constant around the surface. Hence, we can't nicely simplify the problem and the first integral becomes unsolvable (by us).

If the charge falls outside of the Gauss surface, the net flux through the surface is zero.

Mon 02 Feb 2026 11:00

Lecture 5 - Electrostatics V: Gauss Examples and Electric Potential

1 Non-Conducting Infinite Charged Parallel Plates

Consider two non-conducting infinite planes of charge with charge density σ .

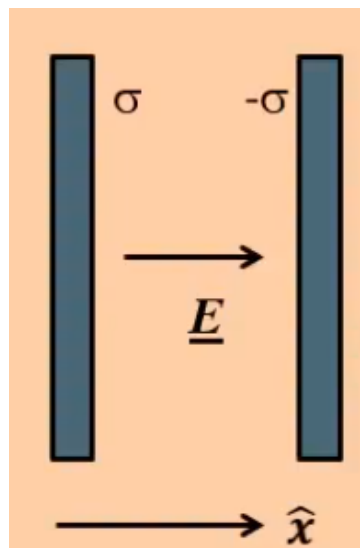


Figure 5.1

We have derived that the surface charge from the LHS plate gives field:

$$\underline{E}_L = \frac{\sigma}{2\epsilon_0} \hat{x}$$

The RHS plate gives:

$$\underline{E}_R = \frac{-\sigma}{2\epsilon_0} \hat{n} = \frac{\sigma}{2\epsilon_0} \hat{x}$$

Where $\hat{n}$ is the unit vector normal to the RHS surface. By superposition, the total field between the plates is:

$$\underline{E} = \underline{E}_L + \underline{E}_R = \frac{\sigma}{\epsilon_0} \hat{x}$$

This is exactly the same as for two conducting plates, per last lecture.

1.1 Back to Conducting Plates

Consider two oppositely charged conducting plates a long way apart. They have charges on each surface.

As the two plates move closer together, their E-fields attract the charges on each other. This causes movement of the charges, until all the charges move towards the front faces. This causes the effective surface charge density to double, hence:

$$E = \frac{\sigma_i}{\epsilon_0} + \frac{\sigma_i}{\epsilon_0} = \frac{2\sigma_i}{\epsilon_0} = \frac{\sigma}{\epsilon_0}$$

Where σ_i is the initial charge densities on the surface of each plate, and σ is the final charge density effective on the inner surfaces. There is therefore no charges (and hence $E = 0$) on the outer surfaces. Note that this is only true when the plates have equal and opposite charge densities.

2 Work Done by an E-Fields

Consider a particle moving in an E-field. Across some infinitesimal distance $\delta \underline{l}$, over which the particle experiences force $\underline{F}$, the infinitesimal element of work done is:

$$\delta W = \underline{F} \cdot \delta \underline{l}$$

As the E-field by definition carries units of force per unit charge, the force is given by:

$$\delta W = q \underline{E} \cdot \delta \underline{L}$$

The work done by an E-field represents a decrease in electric potential energy, $\delta U = -\delta W$:

$$\delta U = -q \underline{E} \cdot d\underline{L}$$

If the charge is positive, δU is negative and energy decreases. If the charge is negative, δU is positive and energy increases.

If the E-field varies wrt space, and the particle moves from point A to point B, we need to integrate to find the total change:

$$U_B - U_A = -q \int_a^b \underline{E} \cdot d\underline{l}$$

We want to be able to define this on a per unit charge basis.

V is the potential energy of the system *per unit charge*.

$$V = \frac{U}{q}$$

$$V_b - V_A = - \int_a^b \underline{E} \cdot d\underline{l}$$

It is a property of a point everywhere in an E-field. **It is a scalar**. The units are Joules per Coulomb, i.e. Volts.

The electric potential at distance r from a point charge q is:

$$\underline{E} = \frac{q}{4\pi\epsilon_0 r^2} \underline{r}$$

$$V(\infty) - V(r) = - \int_{\infty}^r \underline{E} \cdot d\underline{r} = \frac{q}{4\pi\epsilon_0 r}$$

This is the Coulomb potential, which can be positive or negative (as q can be positive or negative).

As it's potential per unit charge, the U of a charge q_0 at distance r from charge q is:

$$U(r) = V(r)q = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r}$$

And:

$$F = - \frac{dU(r)}{dr} = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2}$$

3 Multiple Charges

What if we have multiple charges? We simply sum across them:

$$V = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i}$$

Where r_i is the distance from the i th charge q_i to the point at which we're evaluating the potential.

4 Electric Potential Difference

The electric potential difference (AKA voltage) between two points is:

$$V_B - V_A = \frac{U_B - U_A}{q} = - \int_a^b \underline{E} \cdot d\underline{l}$$

We can use this to get that Newtons per Coulomb and Volts per Metre are dimensionally equivalent units, and we commonly use the latter for describing electric fields.

5 E-Field Lines

The lines of an E-field trace the path that a positive test charge would follow under the action of the electrostatic force. If the positive test charge were released, it would accelerate along the field lines, from high potential to low potential.

We commonly overlay perpendicular (to $\underline{E}$) lines on a field diagram, representing surfaces where V is a constant. These are called equipotentials and are always perpendicular to $\underline{E}$.

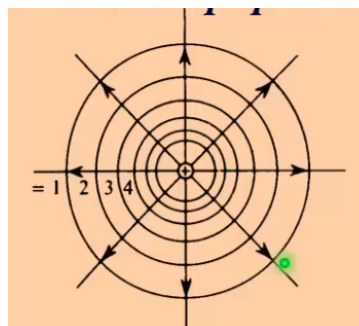


Figure 5.2

6 E from V

Because V is a scalar quantity, its often easier to calculate than E . There are some cases where its therefore easiest to find V and use this to determine E . We start from:

$$V = - \int \underline{E} \cdot d\underline{l}$$

And differentiate both sides:

$$\underline{E} = - \left(\frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k} \right) = - \underline{\nabla} V$$

Or in polar coordinates:

$$\underline{E} = - \left(\frac{\partial V}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{\theta} \right)$$

Where the $1/r$ term arises to ensure dimensional correctness. Often, we have nice examples where we're only dealing with the radial component, so have:

$$E(r) = -\frac{dV}{dr}\hat{r} = \frac{q}{4\pi\epsilon_0 r^2}\hat{r}$$

7 Conservatism

The $\underline{E}$ -field is a conservative field. This means that if the charge returns to its original position, regardless of the route taken, no net work is done:

$$\oint \underline{E} \cdot d\underline{l} = 0$$

This equivalently means that the change in potential between two points is the same whichever path between the two points is taken.

Thu 05 Feb 2026 11:00

Lecture 6 - Electrostatics VI: Electric Potential Examples

1 Potential from Two Charges

Consider two charges aligned on the y-axis:

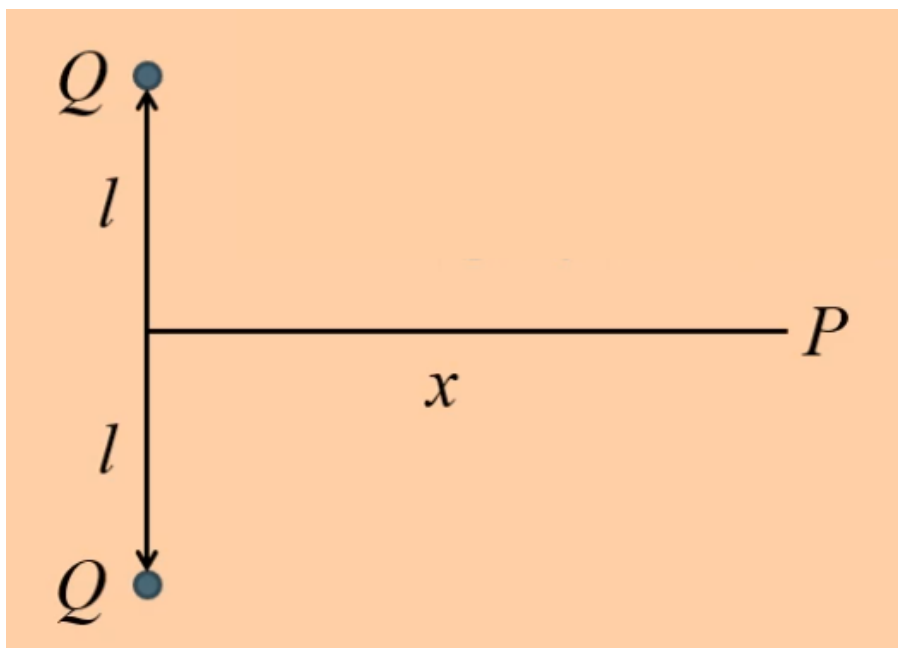


Figure 6.1

What is the Potential and E-field at some point P , and what is the work done moving a charge from horizontal distance x to $2x$?

Let the distance between the charges and P be $r = \sqrt{l^2 + x^2}$. To get the potential at P we can sum the two potentials from Q_1 and Q_2 :

$$V = \frac{Q}{4\pi\epsilon_0 r} + \frac{Q}{4\pi\epsilon_0 r} = \frac{Q}{2\pi\epsilon_0 r}$$

However, the question didn't define r so we need to put it in terms of x and l for a final answer:

$$V = \frac{Q}{2\pi\epsilon_0} \frac{1}{\sqrt{l^2 + x^2}}$$

As we have V , we can get $\mathbf{E}$:

$$\mathbf{E} = -\nabla V = -\frac{\partial V}{\partial x} \hat{x} - \frac{\partial V}{\partial y} \hat{y} - \frac{\partial V}{\partial z} \hat{z}$$

Since our expression for potential is only in terms of x (and by symmetry):

$$\frac{\partial V}{\partial y} = \frac{\partial V}{\partial z} = 0$$

Hence:

$$\begin{aligned}\mathbf{E} &= -\frac{\partial V}{\partial x} \hat{\mathbf{x}} \\ &= -\frac{Q}{2\pi\epsilon_0} \left(-\frac{1}{2} 2x [l^2 + x^2]^{-3/2} \right) \hat{\mathbf{x}} \\ \mathbf{E} &= \frac{Q}{2\pi\epsilon_0} \frac{x}{(l^2 + x^2)^{3/2}} \hat{\mathbf{x}}\end{aligned}$$

And for work done by the E-field moving charge q from $x \rightarrow 2x$, the work done is the change in potential energy.

$$\begin{aligned}\Delta U &= q\Delta V = q(U(2x) - U(x)) \\ \Delta V &= \frac{Q}{2\pi\epsilon_0} \left\{ \frac{1}{(l^2 + 4x^2)^{1/2}} - \frac{1}{(l^2 + x^2)^{1/2}} \right\} \\ \Rightarrow \Delta W &= -q\Delta V = \frac{Qq}{2\pi\epsilon_0} \left\{ \frac{1}{(l^2 + x^2)^{1/2}} - \frac{1}{(l^2 + 4x^2)^{1/2}} \right\}\end{aligned}$$

Note that the negative sign arises because the work is being done by the E-field. If the work was being done on the E-field, the sign would swap.

2 Infinite Charged Line

Consider an infinite line of charge with charge per unit length λ .

From Gauss' law, we have:

$$\begin{aligned}\mathbf{E} &= \frac{\lambda}{2\pi\epsilon_0 r} \hat{\mathbf{r}} \\ V &= -\int \mathbf{E} \cdot d\mathbf{r} = \frac{-\lambda}{2\pi\epsilon_0} \int \frac{dr}{r} \\ &= -\frac{\lambda}{2\pi\epsilon_0} \ln(r) + c\end{aligned}$$

This is a bit problematic, as $V \rightarrow 0$ as $r \rightarrow \infty$..., we leave it in this form. To get around this, we take some finite reference point which we define to have zero potential and normalise to that. The divergence doesn't break anything, because while it's non-physical, the context of an infinite wire is also non-physical.

We can however consider potential differences. Consider two points, A and B .

$$\begin{aligned}\Delta V &= V_B - V_A = \int_a^b \frac{\lambda}{2\pi\epsilon_0} \frac{dr}{r} \\ &= -\frac{\lambda}{2\pi\epsilon_0} \ln \frac{a}{b}\end{aligned}$$

If we wanted to normalise, we'd take b as the finite reference point where potential is zero.

Mon 10 Feb 2026 12:00

Lecture 7 - Electrostatics VII: Electric Dipoles

1 Using Potential to Determine E

Consider two charges Q with distance L between them. The potential at some generic x is:

$$V(x) = \frac{Q}{4\pi\epsilon_0 x} + \frac{Q}{4\pi\epsilon_0(L-x)}$$

$$\mathbf{E} = -\nabla V = -\frac{dV}{dx} \hat{\mathbf{x}}$$

$$= \left(\frac{Q}{4\pi\epsilon_0 x^2} - \frac{Q}{4\pi\epsilon_0(L-x)^2} \right) \hat{\mathbf{x}}$$

At the middle point $x = L/2$:

$$V(x = L/2) = \frac{Q}{4\pi\epsilon_0 L/2} + \frac{Q}{4\pi\epsilon_0(L - L/2)}$$

$$E(x = L/2) = \frac{Q}{4\pi\epsilon_0 L^2/4} - \frac{Q}{4\pi\epsilon_0(L^2/4)} = 0$$

2 Potential from a Ring

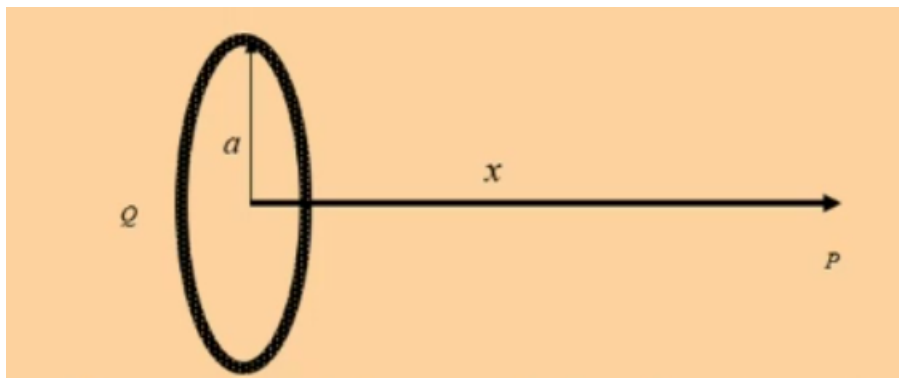


Figure 7.1

A ring-shaped conductor of radius a carries a total charge Q uniformly distributed around it. Find the potential, and hence the electric field, at point P .

Consider some small element of charge δQ , at distance $r = \sqrt{a^2 + x^2}$ from P .

$$\delta V = \frac{\delta Q}{4\pi\epsilon_0 r} = \frac{\delta Q}{4\pi\epsilon_0(a^2 + x^2)^{1/2}}$$

We can then integrate, noting that a, x are both constants:

$$V = \frac{1}{4\pi\epsilon_0(a^2 + x^2)^{1/2}} \oint dQ$$

$$= \frac{Q}{4\pi\epsilon_0(a^2 + x^2)^{1/2}}$$

And then to find $\mathbf{E}$, noting that $\mathbf{E} = -\nabla V$ (and that the derivatives in the z, y directions are zero):

$$\begin{aligned}\mathbf{E} &= -\frac{\partial V}{\partial x} \hat{\mathbf{x}} = -\frac{Q}{4\pi\epsilon_0} \left(-\frac{1}{2} 2x(a^2 + x^2)^{-3/2} \right) \\ &= \frac{Qx}{4\pi\epsilon_0(a^2 + x^2)^{3/2}} \hat{\mathbf{x}}\end{aligned}$$

3 Electric Dipoles

Consider two charges $+q$ and $-q$, separated by a distance a . This forms an electric dipole. This has an associated quantity given the electric dipole moment:

$$\mathbf{p} = q\mathbf{a}$$

We define this vector to point against the $\mathbf{E}$ -field, i.e. pointing from the negative towards the positive charge.

Many molecules naturally form dipoles. Because water is not vertically symmetric, it causes a difference between the effective positive charge from the two hydrogens and the net negative charge from the oxygen.

This makes it an excellent solvent and is used in heating food via microwaves. Because the distance is very small, it gives a very small value of $p = 6.1 \times 10^{-30} \text{Cm}$ per molecule.

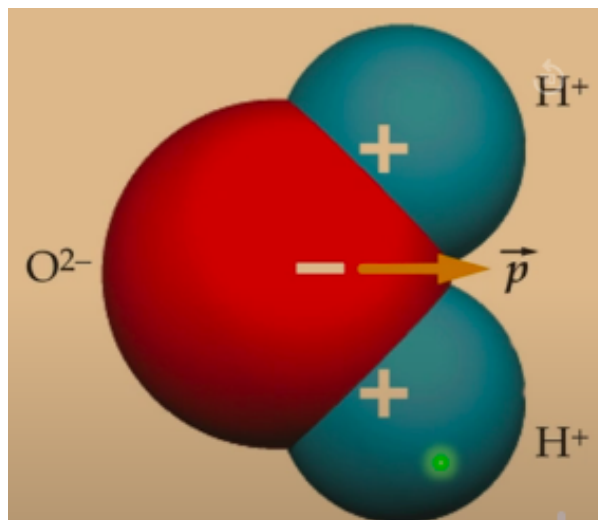


Figure 7.2

For a dipole, the potential is given by:

$$\begin{aligned}V_p &\approx \frac{p \cos \theta}{4\pi\epsilon_0 r^2} \\ \mathbf{E} &\approx \frac{2p \cos \theta}{4\pi\epsilon_0 r^3} \hat{\mathbf{r}} + \frac{p \sin \theta}{4\pi\epsilon_0 r^3} \hat{\boldsymbol{\theta}}\end{aligned}$$

For $r \gg a$ and where θ is defined as the angle between $\mathbf{r}$ and the positive charge.s

Thu 12 Feb 2026 11:00

Lecture 8 - Electrostatics VIII: Electric Dipoles in E-Fields

In this lecture:

- Electric dipoles in uniform external E-fields:
 - Torque
 - Potential energy
 - Work done
 - Right-hand rule
 - Examples

1 Torque

Consider a dipole in a uniform external E-field:

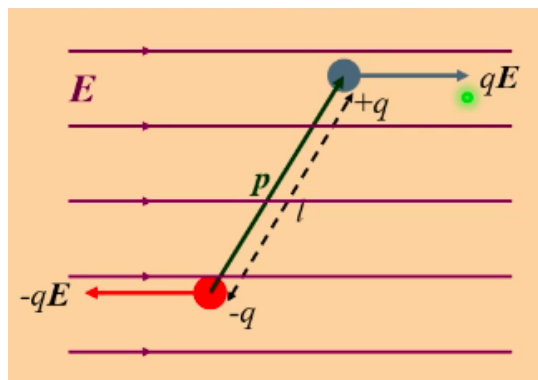


Figure 8.1

As the two charges have equal and opposite forces applied by the field, they do not have a net force. The torque τ rotates the whole couple.

The magnitude of this torque is given by:

$$|\tau| = Fd$$

Where F is one of the two forces and d is the perpendicular distance. This is given by:

$$\tau = |\tau| = Fd = qEd = qEa \sin \theta = pE \sin \theta$$

In vector form it is given as a cross product:

$$\boldsymbol{\tau} = \mathbf{r} \wedge \mathbf{F} = q\mathbf{a} \wedge \mathbf{E} = \mathbf{p} \wedge \mathbf{E}$$

2 Potential Energy

The electric potential at the positive charge be V_+ and hence the electric potential energy is qV_+ , and likewise V_- and qV_- at the negative charge.

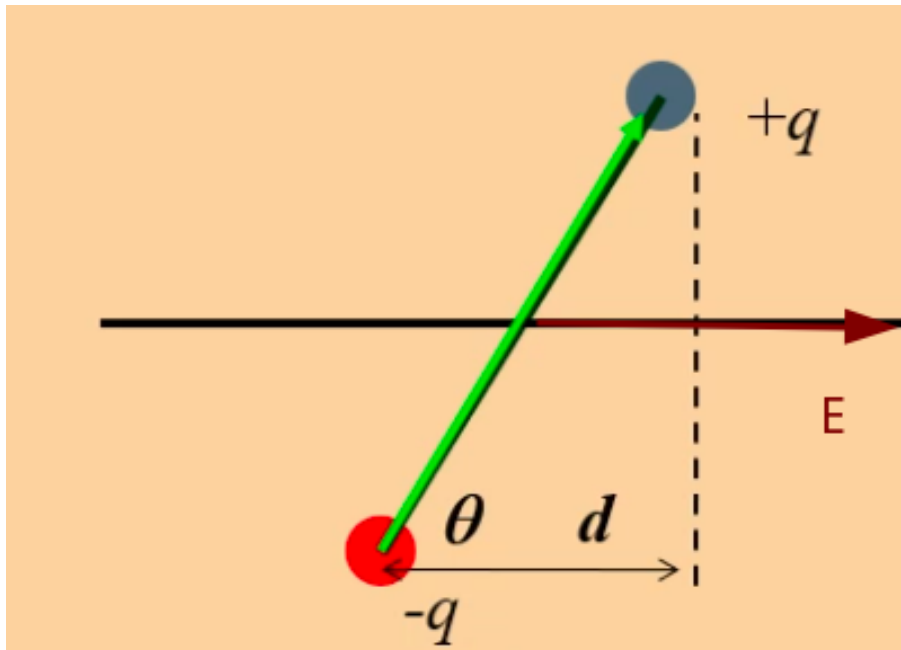


Figure 8.2

The net potential energy on the system is therefore given by:

$$\begin{aligned}
 U &= q(V_+ - V_-) = -q \int_0^d \mathbf{E} \cdot d\mathbf{x} \\
 &= -qEd \\
 &= -qEa \cos \theta = -qaE \cos \theta = -pE \cos \theta
 \end{aligned}$$

Hence in vector notation:

$$U = -\mathbf{p} \cdot \mathbf{E}$$

2.1 Directionality

What is the direction of this setup:

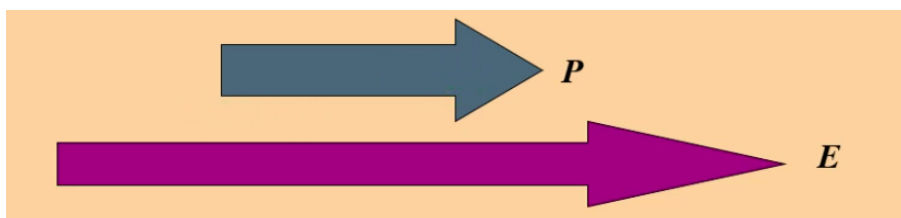


Figure 8.3

Here, as the dipole is aligned with the field, there's no movement. This (as p goes from the negative to positive charge) is the lowest potential energy configuration. The opposite, with p antialigned to the field would still have zero torque, but is the maximum potential energy configuration.

2.2 Microwaves

Dipoles explain how microwave ovens work. For foods containing water, the alternating electric field causes the water molecule's dipole to align itself with the E-field.

Alternating the microwaves cause the water molecules to oscillate in phase. They gain kinetic energy, so heat.

This works for microwaves because a wavelength too low would cause too little a rate of oscillation and a wavelength too high would oscillate the direction of the E-field before the water dipoles have time to fully align. Microwaves fit this region.

3 Work Done

Over some infinitesimal change in angular displacement, $\delta\theta$:

$$\delta W = \tau \delta\theta$$

And takes place in the direction which minimises the angle, hence:

$$\tau = -pE \sin \theta \implies \delta W = -pE \sin \theta \delta\theta$$

So, over a displacement from θ_1 to θ_2 :

$$W = \int dW = \int_{\theta_1}^{\theta_2} -pE \sin \theta d\theta$$

$$W = pE \cos \theta_2 - pE \cos \theta_1 = -(U_2 - U_1) = U_1 - U_2$$

4 Direction of Torque

We can use the Right Hand Rule to determine the direction of the torque:

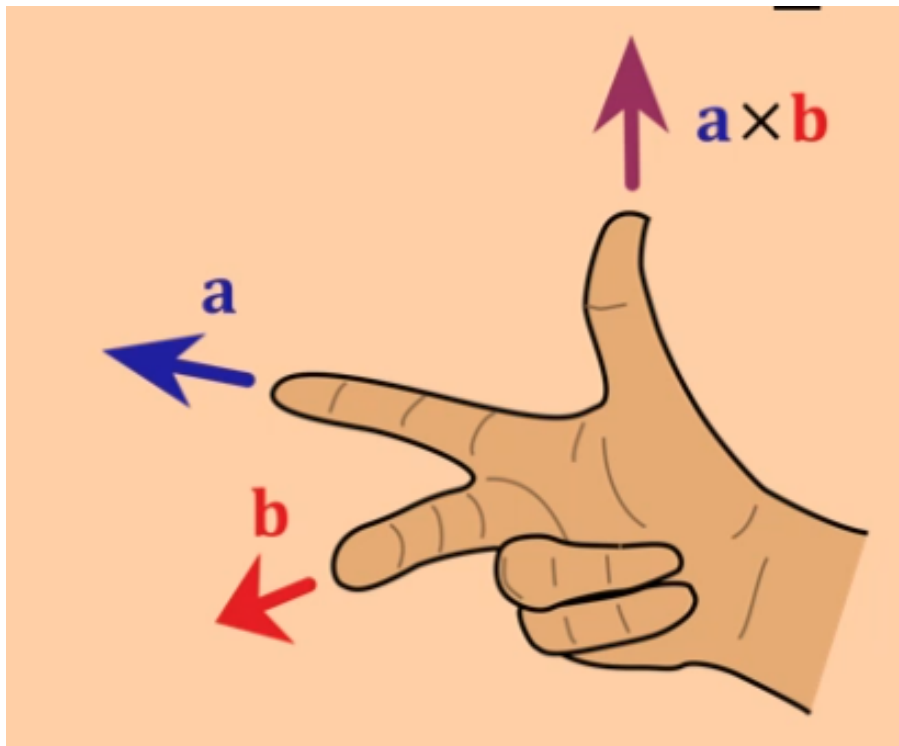


Figure 8.4

Here, **a** represents **p** and **b** represents **E**.

Mon 16 Feb 2026 11:00

Lecture 9 - Electrostatics IX: Capacitance

1 Earthing

Consider two spheres, one of radius a with no charge and a smaller sphere with radius b and charge q .

If we take the two spheres and place them in contact with each other, they will share a potential as some charge from the charged sphere is transferred to the previously uncharged sphere.

This implies that (dropping the $1/4\pi\epsilon_0$ from both sides):

$$\frac{q_a}{a} = \frac{q_b}{b}$$

And the total charge must be conserved, hence:

$$q = q_a + q_b$$

$$\frac{q_a}{a} = \frac{q_b}{b} = \frac{q - q_a}{b} \implies (a + b)q_a = aq$$

Hence:

$$q_a = \frac{aq}{a + b} \approx q \text{ for } a \gg b$$

So, under the same $a \gg b$ condition, $q_b \approx 0$. Effectively, touching a smaller charged sphere to a (much) bigger uncharged sphere distributes the whole charge so finely across the whole system that it is effectively zero.

Formally, earthed isolated bodies share their charge with the earth, and in doing so effectively lose that charge. The bodies and earth acquire a common potential, called the zero of potential. Even though the earth gains some charge, the potential of the earth does not change because a is so large.

2 Capacitance

Playing a charge Q on a conductor alters the conductor's potential by $V \propto Q$. We define the capacitance C of the conductor by the equation:

$$C = \frac{Q}{V}$$

It has units of Coulombs per Volt, aka the Farad. Anything that can hold a charge has an associated capacitance, not just a capacitor.

For example, an isolated conducting sphere of radius R with charge Q has:

$$V = \frac{Q}{4\pi\epsilon_0 R} \implies C = \frac{Q}{V} = 4\pi\epsilon_0 R$$

Capacitance is an inherent property of the material/physical setup, so shouldn't have Q or V in its expression.

In order to get a capacitance of a single farad, you'd need a sphere of $R = 9 \times 10^9 \text{m}$, this is more than 1,000x the that of the earth. Common lab capacitors use picofarads to microfarads.

2.1 Capacitors

A capacitor is a system designed to store charge (therefore energy) by separately storing positive and negative charges.

Suppose some negligible charge δq between the two conductors, whilst they are at a potential difference V_q . The work done here is:

$$\delta W = \delta q V_q$$

We cannot simply integrate this however, as treating V_q as a constant assumes that transferred quantity of charge is negligible and does not impact the overall potential difference. This is not true for non-infinitesimal charges. We cannot therefore say $W = \int_0^Q V_q dq$

We instead rearrange:

$$q = CV_q \implies dq = C dV_q$$

And substitute for dq :

$$W = C \int_0^V V_q dV_q = \frac{1}{2} CV^2$$

This gives the total energy stored in a capacitor:

$$U = \frac{1}{2} CV^2 \quad \left(= \frac{1}{2} QV = \frac{1}{2} \frac{Q^2}{C} \right)$$

3 Calculating Capacitance

The general procedure for determining the capacitance of a system is:

1. Use Gauss' Law to determine the electric field as a function of Q .
2. Integrate to get potential difference.
3. Use $C = Q/V$.
4. Done!

3.1 Parallel Plate Capacitor

We ignore edge effects here and assume we can treat them like infinite plates with a perfectly perpendicular and uniform electric field. This approximation holds where the side lengths of the capacitor are sufficiently larger than the distance between them.

We have two parallel plates, each of area A , with separation d . By Gauss' law:

$$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}$$

As this is constant between the plates, the potential difference is simple to calculate:

$$V = - \int_d^0 \mathbf{E} \cdot d\mathbf{x} = Ed = \frac{qE}{\epsilon_0 A}$$

We integrate against the field to ensure a positively signed answer, but given we only care about a magnitude we could simply do the opposite and drop the negative sign (noting why in an exam).

Therefore:

$$C = \frac{Q}{V} = \frac{\epsilon_0 A}{d}$$

3.2 Long Cylindrical Capacitor

An example is a co-axial cable, with an inner and an outer conductor, separated radially with a dielectric (next lecture) between them:

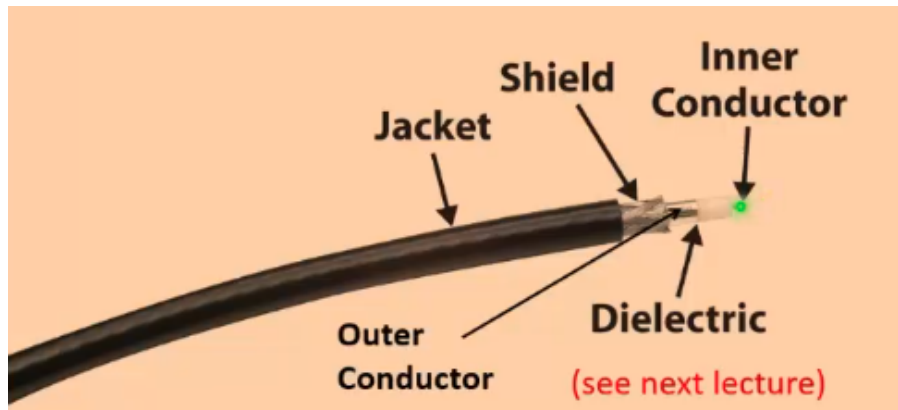


Figure 9.1

Here, we consider some length of the cable L . The inner core has charge per unit length $+\lambda$ and the outer core has $-\lambda$. We want to think about the capacitance per unit length, as in a data transmission cable this is related to the data loss.

We assume the cable is infinite to disregard length effects. The inner core has radius r_a and the outer conductor radius r_b .

We use a cylindrical Gaussian surface of length l and radius R . The RHS of Gauss's law gives:

$$\frac{Q_{\text{enc}}}{\epsilon_0} = \frac{\lambda l}{\epsilon_0}$$

And the electric field is constant and radial ($\perp$ to the Gauss surface) so the LHS is:

$$\int \mathbf{E} \cdot d\mathbf{s} = E 2\pi l$$

Hence:

$$E = \frac{\lambda}{\epsilon_0 2\pi r}$$

$$\mathbf{E} = \frac{\lambda}{\epsilon_0 2\pi r} \hat{\mathbf{r}}$$

Therefore the potential difference is:

$$V = - \int_{r_b}^{r_a} \mathbf{E} \cdot d\mathbf{r} = - \frac{\lambda}{2\pi\epsilon_0} [\ln r]_{r_b}^{r_a}$$

And swapping the bounds to clear the negative and get magnitude only:

$$V = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{r_b}{r_a}\right)$$

Hence capacitance per unit length is:

$$C = \frac{Q}{V} = \frac{\lambda}{V} = \frac{2\pi\epsilon_0}{\ln(r_b/r_a)}$$

3.3 Concentric Spheres

A spherical capacitor has two spherical conductors, an inner sphere with radius r_a and surface charge density $+q$ and an outer sphere with $-q$ and r_b . We place a spherical Gaussian surface concentric to these spheres, and between them.

Applying Gauss' law (as previously done for a sphere) we get:

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r}$$

Hence potential difference is:

$$\begin{aligned} V &= - \int_{r_b}^{r_a} \mathbf{E} \cdot d\mathbf{r} \\ &= - \frac{Q}{4\pi\epsilon_0} \int_{r_b}^{r_a} \frac{1}{r^2} dr = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r_a} - \frac{1}{r_b} \right) \\ &= \frac{Q}{4\pi\epsilon_0} \frac{(r_b - r_a)}{r_a r_b} \\ \Rightarrow \quad C &= \frac{Q}{V} = \frac{4\pi\epsilon_0 r_a r_b}{r_b - r_a} \end{aligned}$$

Note that the single sphere case defined earlier is effectively just a generalisation of this case where the outer sphere goes off to infinity, i.e. $r_b \rightarrow \infty$.

Thu 19 Feb 2026 11:00

Lecture 10 - Electrostatics X: Dielectrics

1 Energy Density

Consider a parallel plate capacitor We have previously determined that:

$$U = \frac{1}{2}CV^2$$

And that:

$$C = \frac{Q}{V} = \frac{\epsilon_0 A}{d}$$

$$V = Ed$$

Hence:

$$U = \frac{1}{2} \frac{\epsilon_0 A}{d} E^2 d^2 = \frac{1}{2} Ad \epsilon_0 E^2$$

Noting that Ad is just the volume of the gap, we have that the energy per unit volume is given by:

$$u_E = \frac{1}{2} \epsilon_0 E^2$$

this gives the energy density of an electric field (for any E-field in a vacuum).

2 Force Per Capacitor Plates

Again for a parallel plate capacitor, we want to find the attractive force between the plates. Integrating coulombs law directly would be overly messy, so we use the definition:

$$U = - \int \mathbf{F} \cdot d\mathbf{x}$$

Where x is the axis between the plates. Hence $F = -du/dx$:

$$U(x) = Ax\epsilon_0 E^2$$

$$F = -\frac{1}{2}A\epsilon_0 E^2$$

Using the definition:

$$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{A\epsilon_0}$$

$$\Rightarrow F = -\frac{1}{2}A\epsilon_0 \left(\frac{Q}{A\epsilon_0} \right)^2$$

$$F = -\frac{1}{2}QE$$

And as the force is negatively signed, that confirm that it is attractive.

3 Dielectrics

What is a dielectric?

In a perfect conductor, there's a large quantity of free charges. Under an applied E-field, these charges move rapidly along the direction of the field.

A dielectric is a poor conductor where free charge is negligible. When an E-field is applied, the charges in the atom are displaced, while remaining bound to the atom. This is called polarisation.

They are placed between the plates of a capacitor to keep the charged surfaces physically separated and to raise the capacitance. For EMI (and EMII) we only consider well behaved dielectrics which follow the acronym "HILS"

- Homogeneous.
- Isotropic, same in all directions hence $P \parallel E$.
- Linear, $P \propto E$.
- Stationary, no moving parts.

4 Polarisation

Polarisation is the effect of an E-field on a dielectric to pull the positively charged nucleus in one direction and the negatively charged electron cloud in the other. The electron cloud and the nucleus remain bound.

The E-field slightly pulls the nucleus in the direction of the E-field and the electron cloud against it. This creates a dipole moment in the direction of the E-field. This polarisation creates an E-field which acts in the opposite direction to the E-field, cancelling a small part of it.

All of these dipoles align and act together oppositely to the E-field. This reduces the field in the electric.

If the dielectric has N dipoles, we define the total polarisation as:

$$\mathbf{P} = N\mathbf{p} = Nq\mathbf{a}$$

The field inside the dielectric is less than the external applied field.

Consider a large dielectric block placed into a field. While the block remains neutral, the polarisation causes a small depletion region, approximately the width of the dipoles at each end of the block (width a). One end of the block becomes depleted of negative charges and one end positive charges.

The charge on the surface (surface area S) is given by:

$$\text{surface charge} = \text{volume of depleted region} \times \text{molecular density} \times \text{charge per molecule} = (aS)Nq$$

However we've already defined $P = Nqa$, so the surface charge on the RHS (for a right to left E-field) is:

$$q_r = PS$$

And the LHS:

$$q_l = -PS$$

Hence the surface charge density is given by $\sigma_i \pm P$.

If we add a dielectric to a capacitor between the plates, the dielectric surface with negative surface charge is in contact with the capacitor plate of positive surface charge. The two E-fields partially cancel and, while the net E-field still points in the same direction (positive plate to negative plate) it has a smaller magnitude.

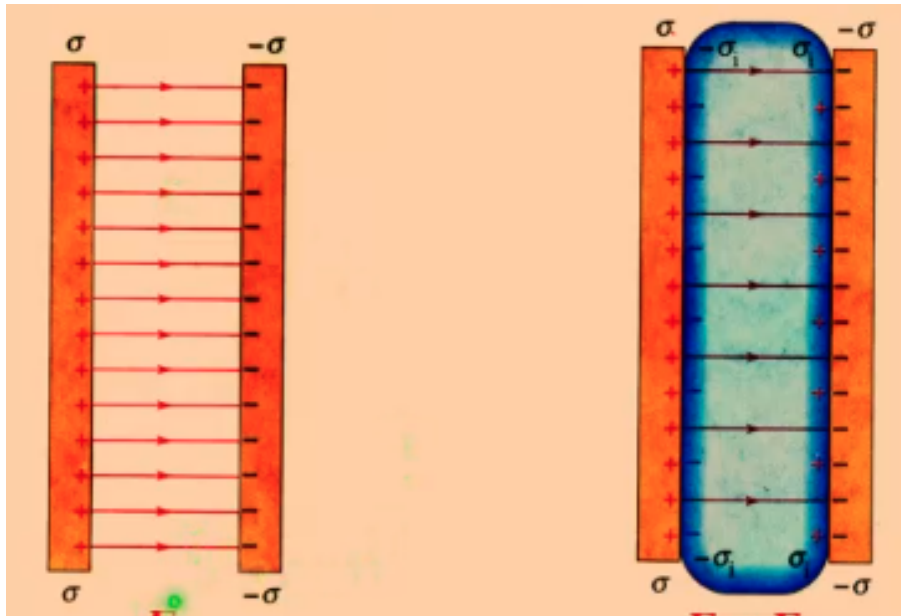


Figure 10.1

Considering the positive plate, the net surface charge is $\sigma - \sigma_i$. With no dielectric, the E-field is:

$$E_0 = \frac{\sigma}{\epsilon_0}$$

With the dielectric:

$$E = \frac{\sigma - \sigma_i}{\epsilon_0} = \frac{\sigma - P}{\epsilon_0}$$

For a well-behaved dielectric, the polarisation is proportional to the electric field $P \propto E$, hence the polarisation must be some constant times the electric field:

$$\mathbf{P} = \chi_E \epsilon_0 \mathbf{E}$$

Where $\chi_E \epsilon_0$ is the constant of proportionality and ϵ_0 is included to make sure that χ_E is dimensionless. χ_E is defined as the susceptibility.

We therefore have:

$$\begin{aligned} \epsilon_0 E &= \sigma - P \\ \epsilon_0 E &= (\epsilon_0 E_0) - (\chi_E \epsilon_0 E) \\ \epsilon_0 E &= \epsilon_0 E_0 - \chi_E \epsilon_0 E \\ \boxed{(1 + \chi_E) \epsilon_0 E} &= \epsilon_0 E_0 \end{aligned}$$

We define another dimensionless constant, the relative permittivity (AKA the dielectric constant):

$$\begin{aligned} \epsilon_r &= (1 + \chi_E) \\ \implies \epsilon_r \epsilon_0 E &= \epsilon_0 E_0 \end{aligned}$$

The E-field in the dielectric is:

$$E = \frac{E_0}{\epsilon_r}$$

From Gauss' law we have:

$$\begin{aligned} E &= \frac{\sigma}{\epsilon_r \epsilon_0} = \frac{Q}{\epsilon_r \epsilon_0 A} \\ V &= V_+ - V_- = Ed = \frac{Qd}{\epsilon_0 \epsilon_r} \\ \implies C &= \frac{Q}{V} = \frac{\epsilon_0 \epsilon_r A}{d} \end{aligned}$$

Since $\epsilon_r > 1$, this increases the capacitance. We can modify Gauss' law for a well behaved dielectric:

$$\int_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q_{\text{enc}}}{\epsilon_r \epsilon_0}$$

Material	ϵ_r
vacuum	1 (by definition)
air	1.00059
PTFR/Teflon	2.1
Polystyrene	2.4–2.7
Mylar	3.1
Concrete	4.5
Pyrex (glass)	4.7 (3.7–10)

Figure 10.2: Typical Values for ϵ_r

Mon 23 Feb 2026 11:00

Lecture 11 - Electrostatics XI: Revision

1 Example I

Two point charges $+Q$ and $-Q$ are at (x, y) coordinates $(-2a, 0)$ and $2a, 0$ respectively.

What is the potential at $(2a, 3a)$? We work out the contributions in potential from each charge and sum them.

For the positive charge:

$$V = \frac{Q}{4\pi\epsilon_0\sqrt{(3a)^2 + (4a)^2}} = \frac{Q}{4\pi\epsilon_0 5a}$$

And the negative charge:

$$V = -\frac{Q}{4\pi\epsilon_0 3a}$$

Hence the total:

$$\begin{aligned} V_T &= \frac{Q}{4\pi\epsilon_0 a} \left[\frac{1}{5} - \frac{1}{3} \right] \\ &= -\frac{Q}{4\pi\epsilon_0 a} \frac{2}{15} \end{aligned}$$

What is the work done to move the charge from $(-2a, 3a)$ to $(-2a, -3a)$?

$$W = q\Delta V$$

Where the initial potential is:

$$\begin{aligned} V(-2a, 3a) &= \frac{Q}{4\pi\epsilon_0(3a)} - \frac{Q}{4\pi\epsilon_0(5a)} \\ &= \frac{Q}{4\pi\epsilon_0 a} \frac{2}{15} \end{aligned}$$

And the final potential is:

$$\begin{aligned} V(-2a, -3a) &= \frac{Q}{4\pi\epsilon_0(5a)} - \frac{Q}{4\pi\epsilon_0(3a)} \\ &= -\frac{Q}{4\pi\epsilon_0 a} \frac{2}{15} \end{aligned}$$

Hence the work is given by:

$$\begin{aligned} W &= q\Delta V = q(V(-2a, 3a) - V(-2a, -3a)) \\ &= q\left(\frac{Q}{4\pi\epsilon_0 a} \frac{2}{15} - \left(-\frac{Q}{4\pi\epsilon_0 a} \frac{2}{15}\right)\right) \\ &= \frac{4qQ}{15 \cdot 4\pi\epsilon_0 a} \\ &= \frac{qQ}{15\pi\epsilon_0 a} \end{aligned}$$

Since we're going from a higher potential to a lower potential, the work done is positive, as expected. This corresponds to work being done *by* the electric field.

2 Example II

An infinitely long hollow cylinder has internal radius R_1 and external radius R_2 . For $R_1 < r < R_2$, the cylinder has charge density $\rho(r) = \rho_0 r$, and for $r < R_1$ and $r > R_2$, $\rho(r) = 0$.

1. Show that the total charge Q per unit length of the cylinder is:

$$Q = \frac{2}{3}\pi\rho_0(R_2^3 - R_1^3)$$

2. Calculate the electric field at radius r for:

- (a) $r > R_2$

- (b) $R_1 < r < R_2$

Expressing in terms of r and ρ_0 .

Sketch a graph to show how the field varies with r .

3. Use the previous result to derive an expression for $V(r)$ for $r < R_2$

Answer 1

Consider a cylindrical shell at radius r and thickness δr . Because the shell has infinitesimal thickness, we assume that charge density is constant throughout.

$$\delta Q = \rho_0 r \times 2\pi r \delta r \times L$$

And at unit length $L = 1$

$$\delta Q = \rho_0 r \times 2\pi r \delta r$$

Hence we can integrate for the total charge:

$$\begin{aligned} Q &= \int_{R_1}^{R_2} \rho_0 2\pi r^2 dr \\ &= 2\pi\rho_0 \left(\frac{R_2^3}{3} - \frac{R_1^3}{3} \right) \end{aligned}$$

Hence:

$$Q = \frac{2}{3}\pi\rho_0(R_2^3 - R_1^3)$$

Answer 2

We use a cylindrical Gaussian surface of radius r . Ignoring edge effects, the electric field is constant on the surface and directed radially outwards. Hence $\vec{E}$ and $d\vec{S}$ are parallel, and we can write:

$$\begin{aligned} \int_S \vec{E} \cdot d\vec{S} &= \frac{Q_{\text{enc}}}{\epsilon_0} \\ ES &= \frac{Q_{\text{enc}}}{\epsilon_0} \end{aligned}$$

Initially taking it outside the cylinder, $r > R_2$:

$$\begin{aligned} \text{LHS} &= 2\pi rLE \\ \text{RHS} &= \frac{1}{\epsilon_0} \frac{2}{3}\pi\rho_0(R_2^3 - R_1^3) \\ \implies E &= \frac{\rho_0}{3\epsilon_0} \frac{R_2^3 - R_1^3}{r} \text{ radially} \end{aligned}$$

For $R_1 < r < R_2$:

$$\text{LHS} = 2\pi rLE$$

The RHS contains a smaller volume of charge, so we integrate up to r instead of R_2 :

$$\begin{aligned}\text{RHS} &= \frac{1}{\epsilon_0} \int_{R_1}^r \rho_0 2\pi r^2 dr \\ &= \frac{2\pi\rho_0 L}{3\epsilon_0} (r^3 - R_1^3)\end{aligned}$$

Hence:

$$\begin{aligned}2\pi r L E &= \frac{2\pi\rho_0 L}{3\epsilon_0} (r^3 - R_1^3) \\ E &= \frac{\rho_0}{3\epsilon_0} \frac{r^3 - R_1^3}{r} = \frac{\rho_0}{3\epsilon_0} \left(r^2 - \frac{R_1^3}{r} \right) \text{ radially}\end{aligned}$$

To sketch, we get:

- No enclosed charge, hence no field for $r < R_1$.
- Cubic growth from $R_1 < r < R_2$.
- $1/r$ decat $r > R_2$.

Answer 3

For $r > R_2$, we have determined:

$$E = \frac{\rho_0}{3\epsilon_0} \frac{R_2^3 - R_1^3}{r}$$

So:

$$\begin{aligned}V &= -\frac{\rho_0}{3\epsilon_0} (R_2^3 - R_1^3) \int \frac{1}{r} dr \\ &= -\frac{\rho_0}{3\epsilon_0} (R_2^3 - R_1^3) \ln r + c\end{aligned}$$

As this doesn't tend to zero as $r \rightarrow \infty$, we cannot determine the constant c . We would need a reference point at some finite r to determine c , but we don't have that information, so we leave it as is.

3 Example III

A wire is bent into a semicircle of radius R . It has uniform charge density $+\lambda$ per unit length. What is the magnitude and direction of the electric field at the centre of the semicircle?

Because it's not symmetric, we can't use Gauss's law and need to use the integration method. We take a small arc length of the wire $\delta\theta$ subtending a charge δq . This creates a field at $\delta\theta$ from the horizontal. For each horizontal component in one direction, there is an equal and opposite component in the opposite direction, so the horizontal components cancel out. Hence by symmetry the resultant direction is the negative vertical and we can only consider magnitude.

$$\delta E_y = \frac{\delta Q}{4\pi\epsilon_0 R^2} = \frac{\lambda\delta L}{4\pi\epsilon_0 R^2}$$

And as the subtended arc length is $\delta L = R\delta\theta$,

$$\delta E = \frac{\lambda R \delta\theta}{4\pi\epsilon_0 R^2}$$

$$\delta E = \frac{\lambda \delta\theta}{4\pi\epsilon_0 R}$$

$$\delta E_y = \delta E \sin \theta$$

$$\delta E_y = \frac{\lambda}{4\pi\epsilon_0 R} \sin \theta \delta\theta$$

And integrating across the semicircle:

$$\begin{aligned}
 E_y &= \int_0^\pi \frac{\lambda}{4\pi\epsilon_0 R} \sin \theta \, d\theta \\
 &= \frac{\lambda}{4\pi\epsilon_0 R} \int_0^\pi \sin \theta \, d\theta \\
 &= \frac{\lambda}{4\pi\epsilon_0 R} [-\cos \theta]_0^\pi \\
 &= \frac{\lambda}{4\pi\epsilon_0 R} (-\cos \pi + \cos 0) \\
 &= \frac{\lambda}{4\pi\epsilon_0 R} (1 + 1) \\
 &= \frac{\lambda}{2\pi\epsilon_0 R}
 \end{aligned}$$

And the direction is vertically downward (negative y-direction).

$$\vec{E} = \frac{-\lambda}{2\pi\epsilon_0 R} \hat{y}$$

4 Example IV

A ring shaped conductor of radius a carries a total charge Q uniformly distributed around it. The ring is in the (y, z) plane, centred on the origin.

1. Find the potential at a point P on the ring axis at a distance x from the centre of the ring.
2. Hence show that the only non-zero component of the E-field at P is along the ring axis and is given by:

$$E_x = \frac{Qx}{4\pi\epsilon_0(a^2 + x^2)^{3/2}}$$

3. If an electron is placed at the centre of the ring and is displaced a small distance x where $x \ll a$. Show that the electron oscillates with a frequency f .

Answer 1

Let $r = \sqrt{a^2 + x^2}$.

$$\begin{aligned}
 V &= \frac{1}{4\pi\epsilon_0 r^2} \oint dQ \\
 &= \frac{1}{4\pi\epsilon_0 r^2} Q \\
 &= \frac{Q}{4\pi\epsilon_0 \sqrt{a^2 + x^2}}
 \end{aligned}$$

Answer 2 Using $E = -\nabla V$:

$$\begin{aligned}
 \vec{E} &= -\nabla V = -\frac{\partial V}{\partial x} \hat{x} - \frac{\partial V}{\partial y} \hat{y} - \frac{\partial V}{\partial z} \hat{z} \\
 \frac{\partial V}{\partial y} &= \frac{\partial V}{\partial z} = 0
 \end{aligned}$$

Hence there is no field in the y and z directions, and the only non-zero component is along the x -axis. $E_y = E_z = 0$.

So:

$$\vec{E} = -\frac{\partial}{\partial x} \left(\frac{Q}{4\pi\epsilon_0 \sqrt{a^2 + x^2}} \right) \hat{x}$$

$$\vec{E} = \frac{Q}{4\pi\epsilon_0} \left(-\frac{1}{2} \times 2x(a^2 + x^2)^{-3/2} \right)$$

$$\vec{E} = \frac{Qx}{4\pi\epsilon_0(a^2 + x^2)^{3/2}} \hat{x}$$

Answer 3

We can rewrite:

$$E = \frac{Q}{4\pi\epsilon_0} \frac{x}{a^3 \left(1 + \frac{x^2}{a^2} \right)^{3/2}}$$

For $x \ll a$, we can approximate ¹:

$$E \approx \frac{Qx}{4\pi\epsilon_0 a^3}$$

And the force is:

$$\vec{F} = -e\vec{E} = \frac{-eQx}{4\pi\epsilon_0 a^3} \hat{x}$$

Using $F = ma$:

$$m \frac{d^2x}{dt^2} = \frac{-eQx}{4\pi\epsilon_0 a^3}$$

$$\frac{d^2x}{dt^2} + \frac{eQ}{4\pi\epsilon_0 ma^3} x = 0$$

This is the equation of a simple harmonic oscillator:

$$x(t) = x_0 \cos \omega t$$

Where:

$$\omega^2 = \frac{eQ}{4\pi\epsilon_0 ma^3}$$

$$\omega = \sqrt{\frac{eQ}{4\pi\epsilon_0 ma^3}}$$

And finally $\omega = 2\pi f$:

$$f = \frac{1}{2\pi} \sqrt{\frac{eQ}{4\pi\epsilon_0 ma^3}}$$

¹Note that if the x term wasn't present in the numerator, we'd have to do a first-order binomial expansion, but as x is present, we can just approximate the denominator to a^3 .

Thu 26 Feb 2026 11:00

Lecture 12 - Magnetism I: Currents and Magnetic Force

Today we start Part II, Magnetism:

- Definition of current and current density.
- Magnetic force on a moving charge.
- The Lorentz Force.
- Field lines for a magnetic field.

1 Current

1.1 Definition of Current

Suppose a current carries a current I , this is defined in terms of the rate of flow of charge past a given cross section:

$$I = \frac{dQ}{dt}$$

The SI unit of current is the *ampere*, defined as the flow of one coulomb per second.

1.2 Nature of Current

There's two experimentally observed effects of a current flow:

- Heating
- Creation of magnetic fields.

Consider a current through a conductor. This is caused by electrons moving (with average drift velocity v) as a result of the induced E-field.

In some time, they travel $v\Delta t$ through the cross section, and if this cross section has area A they sweep out an area of $Av\Delta t$.

If a unit volume has n conducting free electrons, the number of charges is given by:

$$N = n(Av\Delta t)$$

And hence the total charge is:

$$\Delta Q = nAv\Delta t(-e)$$

Hence:

$$\hat{i} = \frac{\Delta Q}{\Delta t} = -nAev$$

1.3 Current Density

More usefully, we talk about the current per unit cross-section, which is the current density J :

$$\hat{j} = \frac{\hat{i}}{A} = -nev$$

The negative sign arises as we define current as flowing from positive to negative. Electrons, being negatively charged, actually physically flow from negative to positive. Current flow and actual electron flow are therefore in the opposite direction. This is because current was initially understood and defined before the electron was defined.

Typically, $v < 1\text{mms}^{-1}$ when a current is flowing.

2 Magnetism

2.1 Magnetic Force on a Charge

A magnetic field exerts a force on a moving charge that is present in the magnetic field.

Suppose a particle of charge $+q$ moving with some velocity $\underline{v}$ in a magnetic field $\underline{B}$ experiences some force $\underline{F}_m$.

We cannot derive from first principles, but experimentally have observed:

- $\underline{F}_m \perp \underline{v}, \underline{B}$.
- $\underline{F}_m \propto \underline{v}$.
- $\underline{F}_m \propto \underline{B}$.
- $\underline{F}_m \propto q$

If θ is the angle between the velocity and the B-field direction, we have:

$$\underline{F}_m = Bqv \sin \theta$$

In vector form:

$$\underline{F}_m = q\underline{v} \wedge \underline{B}$$

Here, $\underline{B}$ has the unit Tesla, T. In base units this is $\text{NC}^{-1}\text{m}^{-1}\text{s}$. A one Tesla magnetic field is really quite powerful, so we also define the Gauss, $1\text{G} = 10^{-4}\text{T}$ to move to a nicer scale.

By definition, if a 1C charge moving at 1m/s perpendicular to a magnetic field experiences a force 1N, the field is 1T.

- Earth's magnetic field is $\sim 0.5\text{G}$
- Poles of a large electromagnetic: $\sim 2\text{T}$.
- Surface of a neutron star: $\sim 10^8\text{T}$.
- The maximum pulsed magnetic field that can create in a lab: $\sim 50\text{T}$.

The Earth's field is believed to be generated by electron currents in the iron alloys in its core. It's not completely understood yet, but convection currents causing these conductive alloys to flow and move is the working theory. This movement causes the North and South poles to swap places on average every 300,000 years.

2.2 The Lorentz Force

In a region where we have both an E-field and a B-field, we have the total force as the vector sum of both:

$$\underline{F} = q(\underline{E} + \underline{v} \wedge \underline{B})$$

This is called the Lorentz force.

Suppose Earth's magnetic field is given by:

$$\underline{B} = B \cos 70^\circ \hat{j} - B \sin 70^\circ \hat{k} \quad \text{where: } B = 5 \times 10^{-5}\text{T}$$

A proton moves in this field with:

$$\underline{v} = 10^7 \hat{j} \text{ms}^{-1}$$

$$\begin{aligned}
 \underline{F}_m &= +e\underline{v} \wedge \underline{B} = 1.6 \times 10^{19} \times 10^7 \times \det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 0 \\ 0 & B_y & B_z \end{bmatrix} \\
 &= 1.6 \times 10^{-12} [\hat{i}(B_z - 0) + \hat{j}(0 - 0) + \hat{k}(0 - 0)] \\
 &= 1.6 \times 10^{-12} B_z \hat{i} \\
 &= -1.6 \times 10^{-12} \times 5 \times 10^{-5} \sin 70^\circ \hat{i} \\
 &= -7.5 \times 10^{-17} \hat{i} \text{N}
 \end{aligned}$$

2.3 Magnetic Field Lines

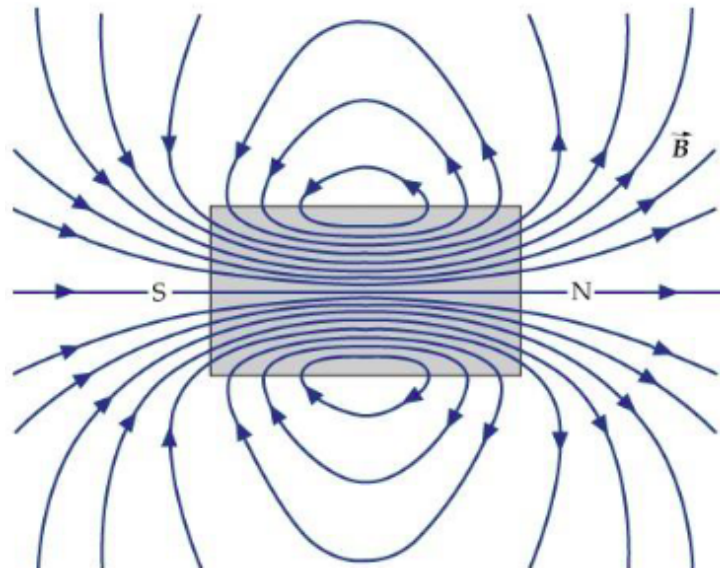


Figure 12.1: Magnetic Field Lines Around a Bar Magnetic

Magnetic field lines are different to electrical field lines. They do not point in the direction of force and travel out of North poles and into South poles. Magnetic field lines are always continuous so there is no magnetic monopoles.

Physicists have searched for a potential magnetic monopoles, and the current upper limit for the number of magnetic monopoles per body is $< 10^{-29}$.

The tangent to a field line at some point gives the direction of B at that point. The number of field lines drawn per unit cross section $\propto B$.

2.4 Magnetic Flux

The magnetic flux $\Delta\phi_B$ has the same definition as electric flux in an E-field. For a small area ΔA :

$$\Delta\phi_B = B \cos \theta \Delta A$$

Where θ is the angle between the B-field and the surface normal vector.

The total magnetic flux is therefore the integral of this:

$$\phi_B = \int_S \underline{B} \cdot d\underline{S}$$

As there is no monopoles, and exiting flux must also return. Therefore, the net flux over any enclosed surface is zero:

$$\int_S \underline{B} \cdot d\underline{S} = 0$$

While this does form one of Maxwell's equations and is very useful for EMII next year, it's not useful in EMI...

Mon 02 Mar 2026 11:00

Lecture 13 - Magnetism II: Magnetic Force and Dipoles

Last lecture:

- Definition of current.
- Current density.
- Magnetic force on a moving charge.
- Lorentz force: $\vec{F} = q(\vec{E} + \vec{v} \wedge \vec{B})$

This lecture:

- Special cases of the magnetic force.
- Force on a current carrying conductor.
- Current loops and magnetic dipoles.
 - Torque on an electric dipole in a B-field.
 - Potential energy of a magnetic dipole in a B-field.

1 Direction of B-Field

- When a B-field is going into the page, we represent it with crosses $\otimes$.
- When a B-field is coming out of the page, we represent it with dots $\odot$.
- This is like the back/front of a dart, when the dart is going into the page we see the cross of feathers, and when the dart is coming out of the page we see the dot of the tip.

1.1 $\vec{V} \perp \vec{B}$

When the velocity of the charge is perpendicular to the magnetic field, the force is given by:

$$F = qvB$$

This force, perpendicular to both the velocity and the magnetic field, acts as a centripetal force, and hence the charge moves in a circle. We can equate the magnetic force to the general centripetal force for circular motion:

$$qvB = \frac{mv^2}{r}$$

$$\therefore r = \frac{mv}{qB} = \frac{p}{qB}$$

This gives us the radius of the circular motion, and this holds for relativistic particles too, as a γ factor would appear in both the denominator and numerator, and hence cancel out.

This also gives us:

$$\frac{qB}{m} = \frac{v}{r} = \omega$$

Where ω is the angular velocity of the circular motion. Hence the frequency is:

$$f = \frac{\omega}{2\pi} = \frac{qB}{2\pi m}$$

1.2 $\vec{v}$ makes an angle θ with $\vec{B}$

When the velocity of the charge makes an angle θ with the magnetic field, we can resolve the velocity into two components $v \sin \theta$ and $v \cos \theta$. The $v \sin \theta$ component contributes to the magnetic force, while the other component moves the charge along the magnetic field.

This gives a helical path:

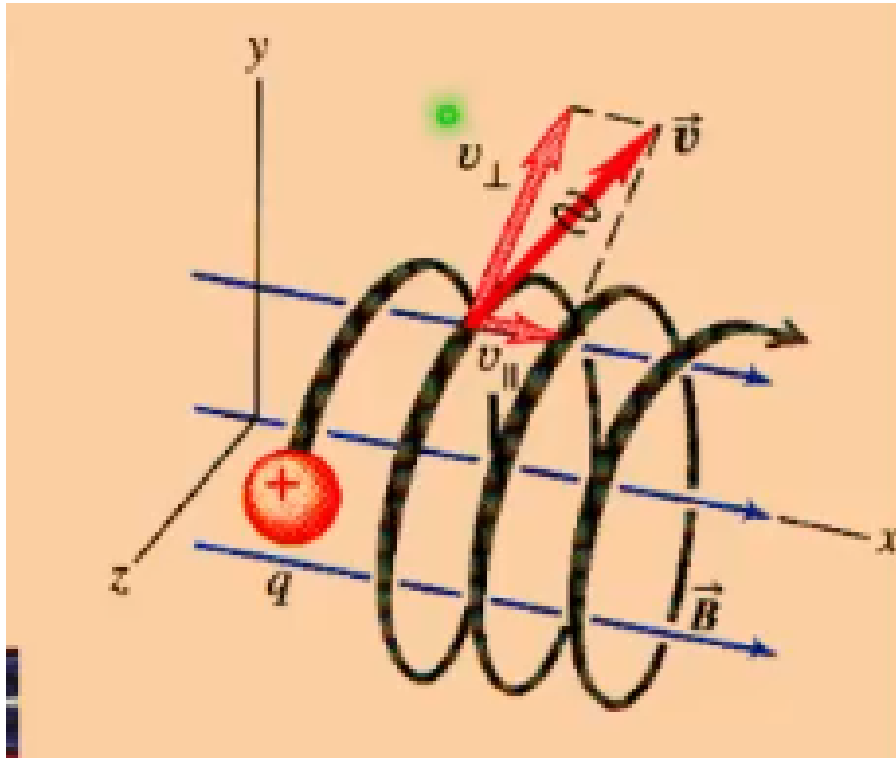


Figure 13.1

Along the x-direction, the motion is clockwise for a negative charge and anticlockwise for positive.

The “cyclotron angular frequency” is given by:

$$\omega = \frac{qB}{m}$$

2 Force on a Current Carrying Conductor

We have established that the magnetic force on a single charge is:

$$\vec{F} = q\vec{v} \wedge \vec{B}$$

If the conductor has cross sectional area A , length l and N charge carriers, give by:

$$N = nAl$$

Where n is the number of charge carriers per unit volume. The total force is:

$$F = Nq\vec{v} \wedge \vec{B} = -nAlq\vec{v} \wedge \vec{B}$$

We’ve defined current as $I = nqAv$, hence we can write the force as:

$$F = I\vec{l} \wedge \vec{B}$$

Conventionally, however, we write the length as a vector $\vec{l}$, which points in the direction of the current, and has magnitude equal to the length of the conductor. We therefore consider the magnitude of current:

$$F = I\vec{l} \wedge \vec{B}$$

For currents (and only currents) we can use the left hand rule “FBI”. The thumb points in the direction of the force, the index finger points in the direction of the magnetic field, and the middle finger points in the direction of the current.

3 Force on a Non-Straight Wire

If the conductor isn't straight, we consider individual length elements of the conductor:

$$\delta \vec{F} = I \delta \vec{l} \wedge \vec{B}$$

The total force is given by integrating:

$$\vec{F} = I \int_a^b d\vec{l} \wedge \vec{B}$$

4 Current Loops

A current loop is known as a magnetic dipole. The magnetude of the magnetic dipole moment is given by:

$$\mu = I \times A$$

Where A is the area of the loop. To get a direction, we express the area as a vector $\vec{A} = A\vec{n}$, where $\vec{n}$ is a unit vector perpendicular to the plane of the loop. The direction of $\vec{n}$ is given by the right hand rule: if the fingers of the right hand curl in the direction of the current, then the thumb points in $\vec{n}$.

Hence:

$$\vec{\mu} = I\vec{A}$$

With units of Am^2 . A current loop produces a magnetic field similar to that of a bar magnetic.

If the current loop is in a B-field, there is no net torque or force.

Consider a current loop at an angle in a B-field. For a loop of any shape with area A in a uniform B-field, the torque is given by:

$$\vec{\tau} = I\vec{A} \wedge \vec{B}$$

Also, since the magnetic dipole moment is defined as $\vec{\mu} = I\vec{A}$, the torque can be given as:

$$\vec{\tau} = \vec{\mu} \wedge \vec{B}$$

5 Potential Energy of a Dipole

Consider a current loop at angle θ in a B-field. The work done by moving this dipole a small angle is:

$$\delta W = \tau \delta \theta$$

This torque is in the direction of decreasing theta, hence:

Thu 05 Mar 2026 11:00

Lecture 14 - Magnetism III: Magnetic Fields from Currents

Recap of last lecture:

- Special cases of the magnetic force.
- Force on a current carrying conductor:

$$\underline{F} = I \underline{L} \wedge \underline{B}$$

- Current loops and magnetic dipoles:

$$\underline{\mu} = I \underline{A}$$

- Torque on a magnetic dipole in a B-field

$$\underline{\tau} = \underline{\mu} \wedge \underline{B}$$

- Potential energy of a magnetic dipole in a B-field:

$$U = -\underline{\mu} \cdot \underline{B}$$

1 Magnetic Fields from a Moving Charge (Current)

Danish physicist Hans Christian Oersted in 1820 demonstrated that a magnetic field exists around a current-carrying wire.

This was the first connection between electric and magnetic phenomena.

Consider a B-field at point $\underline{r}$ from a charge q , moving with velocity $\underline{v}$. The angle between $\underline{r}$ and $\underline{v}$ is given by ϕ .

The induced B-field is perpendicular to both $\underline{u}$ and $\underline{r}$ and is given by:

$$B \propto \frac{qv \sin \phi}{r^2}$$

In vector form, we have:

$$\underline{B} \propto \frac{q}{r^2} \underline{u} \wedge \hat{r}$$

In SI units, the constant of proportionality (involving “permeability of free space”, μ_0) is such:

$$\underline{B} = \frac{\mu_0}{4\pi} \frac{q}{r^2} \underline{u} \wedge \hat{r} \quad \mu_0 = 4\pi \times 10^{-7} \text{TmA}^{-1}$$

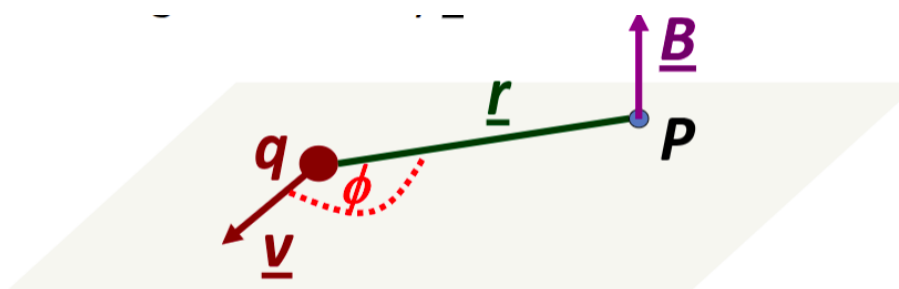


Figure 14.1

In a more complex 3D case, this looks as such:

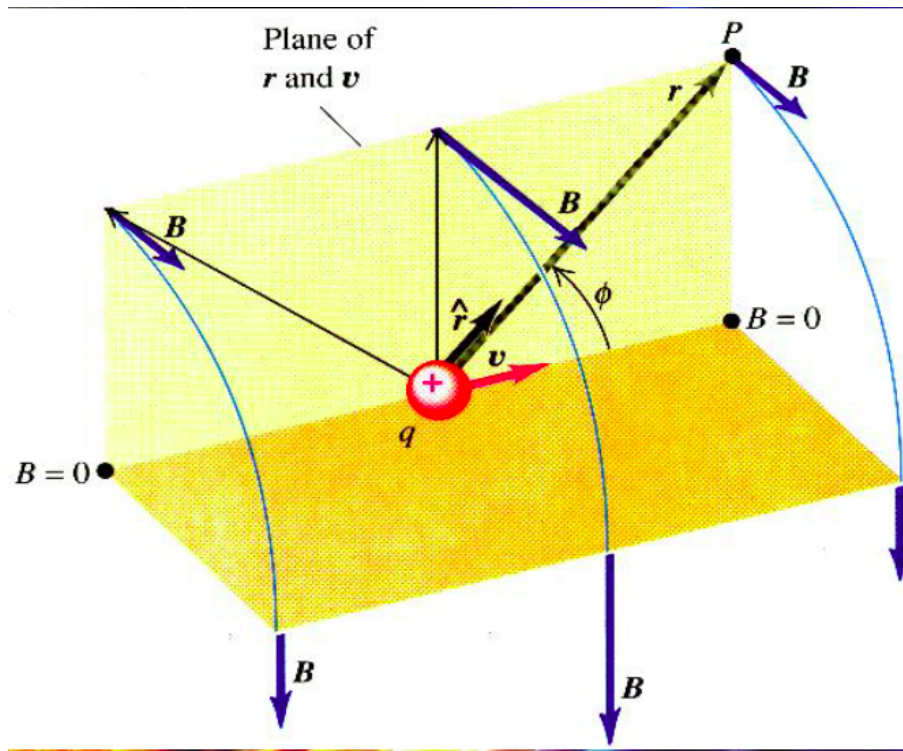


Figure 14.2

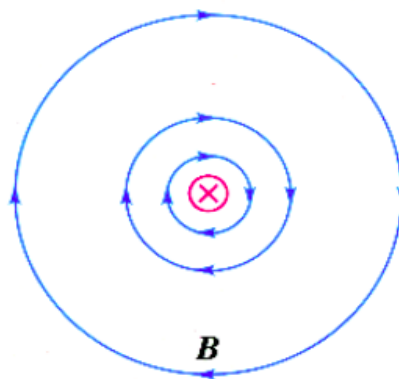


Figure 14.3

The B-field sweeps out a screw shape around the motion of direction. The direction is given by the right hand rule, where a “thumbs up” with the thumb pointing in the direction of the motion of the charge indicates the direction with the curl of the fingers.

1.1 Currents

Consider a wire with a current flow through it. The total charge flowing through some infinitesimal element in a unit time is:

$$I = \frac{qv}{\delta L}$$

Hence:

$$qv = \frac{qv}{\delta L} \delta L = I \delta L \quad \text{where } \delta L \text{ is in the direction of current.}$$

Plugging this in, we get the Biot-Savat Law:

$$\delta \underline{B} = \frac{\mu_0}{4\pi} \frac{I \delta \underline{L} \wedge \hat{r}}{r^2}$$

We can split our object into infinitely many infinitesimal elements and integrate across them all to find the total B-field. We want to:

- Define $d\mathbf{B}$ in terms of a single variable.
- Integrate between the relevant limits.
- Be careful about the directions of the included vector quantities.
- Use symmetry where possible.

2 Current Loops

Consider a current loop with a clockwise current flow. The centre of the loop's circle is P , with a radius from P of R .

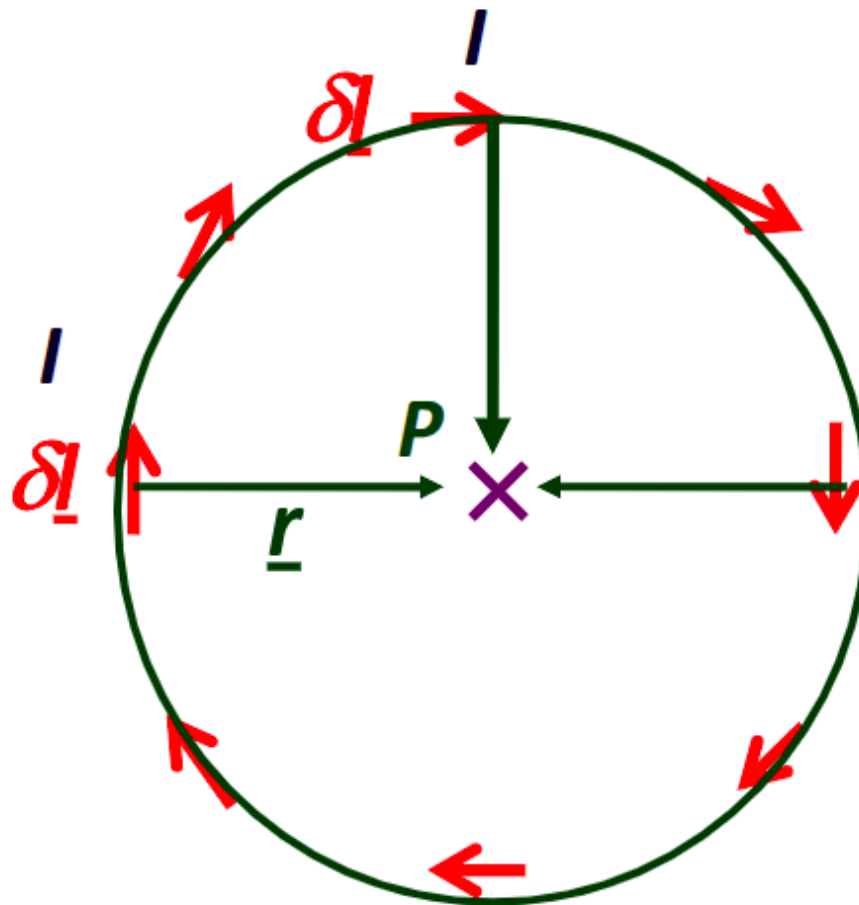


Figure 14.4

The Biot-Savat law gives:

$$\begin{aligned}\delta\mathbf{B} &= \frac{\mu_0}{4\pi} I \frac{\delta\mathbf{l} \wedge \hat{\mathbf{r}}}{r^2} \\ \delta B &= \frac{\mu_0}{4\pi} I \frac{\delta L}{r^2} \\ B &= \frac{\mu_0}{4\pi} \frac{I}{r^2} \oint dL = \frac{\mu_0}{4\pi} \frac{I}{4\pi} \times 2\pi r \\ B &= \frac{\mu_0 I}{2r}\end{aligned}$$

3 Finite Wire

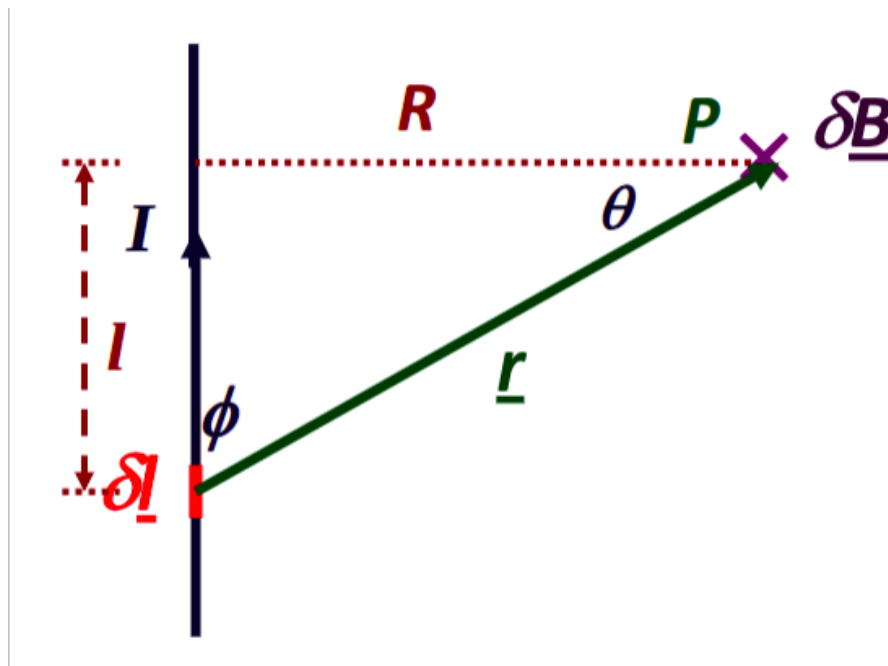


Figure 14.5

We want to find the B-field from the line of current. The B-field is going into the page at P, given by the cross. The Biot Savat Law again gives:

$$\underline{\delta B} = \frac{\mu_0 I}{4\pi r^2} \underline{\delta L} \wedge \hat{r}$$

As $\hat{r}$ is a unit vector, the magnitude is given with the magnitude of L and the sine of phi:

$$\delta B = \frac{\mu_0 I}{4\pi r^2} \delta L \sin \phi \quad (1)$$

We want this in terms of θ instead, and given these two angles form a right angled triangle we have:

$$\sin \phi = \sin \left(\frac{\pi}{2} - \theta \right) = \cos \theta \quad (2)$$

Hence:

$$\delta B = \frac{\mu_0}{4\pi r^2} \delta L \cos \theta$$

By trig we simply have:

$$r \cos \theta = R \implies r = \frac{R}{\cos \theta} \quad (3)$$

$$\begin{aligned} \tan \theta = \frac{L}{R} &\implies L = R \tan \theta \implies \frac{dL}{d\theta} = \frac{R}{\cos^2 \theta} \\ &\implies \delta L = \frac{R}{\cos^2 \theta} \delta \theta \end{aligned} \quad (4)$$

Inserting (2), (3), (4) back into (1) gives:

$$\delta B = \frac{\mu_0 I}{4\pi} \times \frac{\cos^2 \theta}{R^2} \times \cos \theta \times \frac{R}{\cos^2 \theta} \delta \theta$$

Hence:

$$\delta B = \frac{\mu_0 I}{4\pi} \frac{1}{R} \cos \theta \delta \theta$$

$$B = \frac{\mu_0 I}{4\pi R} \int_{\theta_1}^{\theta_2} \cos \theta d\theta$$

Where θ_1, θ_2 depend the angles given by the length of the wire.

$$B = \frac{\mu_0 I}{4\pi R} (\sin \theta_2 - \sin \theta_1)$$

If the wire was infinitely long, $\theta_2 = \pi/2$ and $\theta_1 = -\pi/2$ (technically a limit as $L \rightarrow \infty$). This gives us:

$$B = \frac{\mu_0 I}{2\pi R}$$

4 More Complex Loops

Consider a current loop, with a set of axes centred at the middle of the loop. The loop sits in the yz plane, and we step out some perpendicular distance x in the x plane.

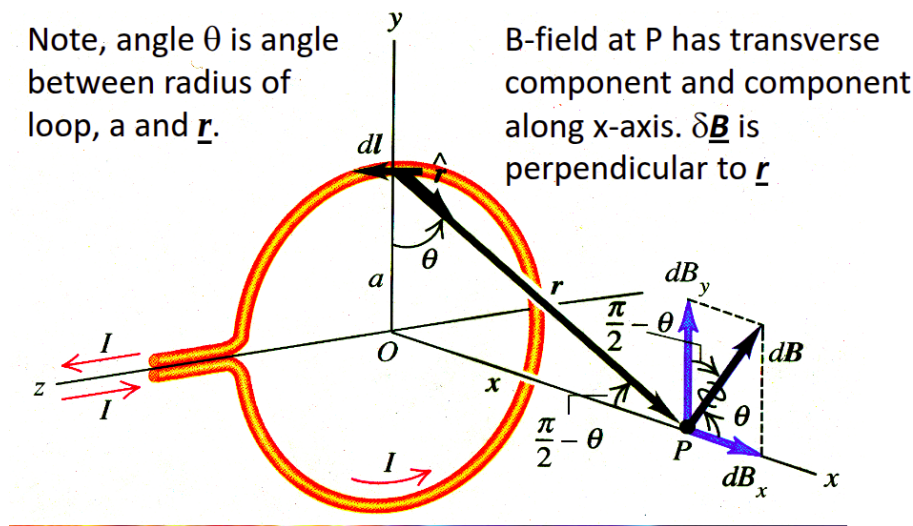


Figure 14.6

We step some distance δL along this loop, and the vector from the point we're investigating to B-field at from this δL is $\underline{r}$, at angle θ .

We can consider symmetry to see that transverse components to the axis will cancel each other out and therefore do not need considering:

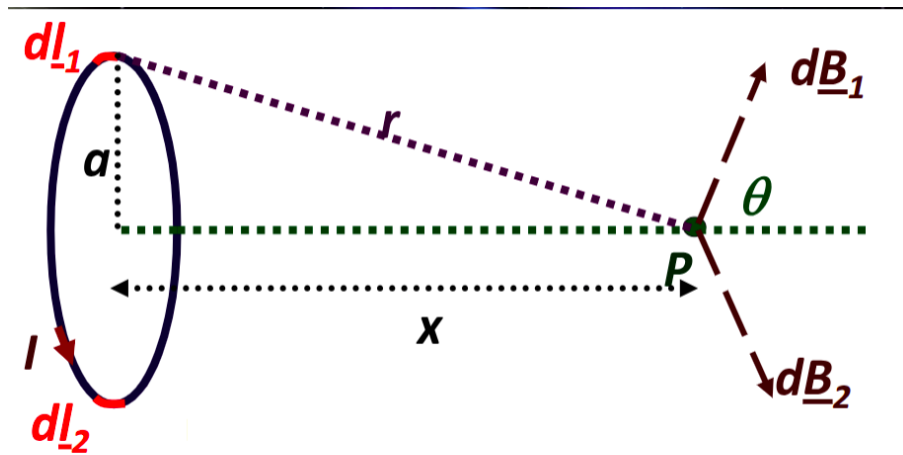


Figure 14.7

Therefore all of the components of the B-field in the yz planes will cancel out, and the only resultant B-field is in the x -direction. We will fully evaluate this on Monday.

Mon 09 Mar 2026 11:00

Lecture 15 - Magnetism IV: Ampere's Law

1 Biot-Savat: Complex Loops Continued

Continuing from the end of last lecture, we found by symmetry that the yz components cancel, leaving only an x component for any point P along the x -axis.

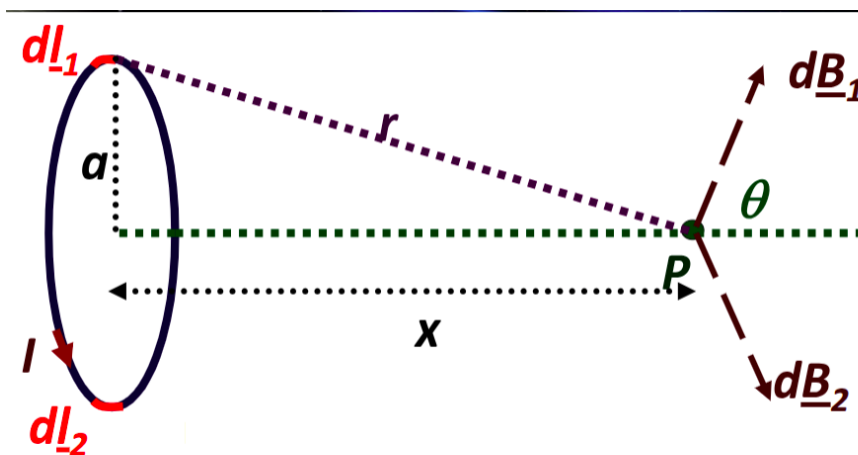


Figure 15.1

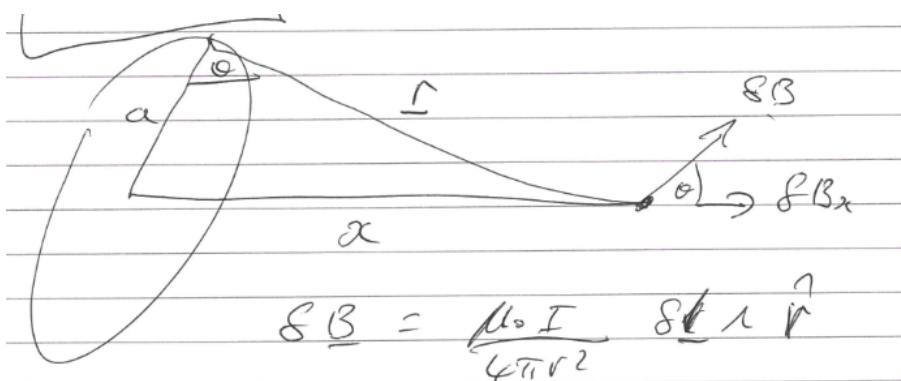


Figure 15.2

Taking magnitudes (and noting that $\hat{r}$ is a unit vector):

$$\delta B = \frac{\mu_0 I}{4\pi r^2} \delta l$$

Resolving into the x -direction, we get that:

$$\delta B_x = \frac{\mu_0 I}{4\pi r^2} \delta l \cos \theta \quad (1)$$

And as we have a right angled triangle:

$$r^2 = a^2 + x^2 \quad (2)$$

$$\cos \theta = \frac{a}{r} \implies \cos \theta = \frac{a}{(a^2 + x^2)^{(1/2)}} \quad (3)$$

Inserting 2 and 3 into 1:

$$\begin{aligned} \delta B_x &= \frac{\mu_0 I}{4\pi} \times \frac{a}{(a^2 + x^2)^{(3/2)}} \delta l \\ B_x &= \frac{\mu_0 I}{4\pi} \times \frac{a}{(a^2 + x^2)^{(3/2)}} \oint \delta l \\ B_x &= \frac{\mu_0 I}{4\pi} \times \frac{a}{(a^2 + x^2)^{(3/2)}} 2\pi a \\ B_x &= \frac{\mu_0 I}{2} \frac{a^2}{(a^2 + x^2)^{(3/2)}} \end{aligned}$$

Taking limits, we can observe that at the centre of the loop (where $x = 0$):

$$B_0 = \frac{\mu_0 I}{2} \frac{a^2}{(a^2)^{(3/2)}} = \frac{\mu_0 I}{2a}$$

And when $x \gg a$, we have:

$$B_x = \frac{\mu_0 I a^2}{2x^3}$$

However, we can simplify using the magnetic dipole moment $\mu_m = I \times \text{area} = I\pi a^2$, hence:

$$B_x = \frac{\mu_0}{4\pi} \frac{2\mu_m}{x^3}$$

This is strikingly similar for the expression for an electric dipole:

$$E_x = \frac{1}{4\pi\epsilon_0} \frac{2p}{x^3}$$

2 E- and B-Fields

We have already determined the following, for E- and B-fields respectively:

$$\int_S \underline{E} \cdot d\underline{S} = \frac{Q}{\epsilon_0} \quad (E)$$

$$\int_S \underline{B} \cdot d\underline{S} = 0 \quad (B)$$

We also know that the E-field is conservative, hence no work is done if a charge returns to its original position:

$$\oint \underline{E} \cdot d\underline{l} = 0$$

Is there an equivalent of this for B-fields?

Consider a circular path of a B-field created around an infinite line of current at radial distance r . By symmetry, $\underline{B}$ and $d\underline{l}$ are parallel, and are constant for a fixed r . Hence:

$$\oint \underline{B} \cdot d\underline{l} = \oint B dl = B \oint dL = B 2\pi r$$

We have also determined that:

$$B = \frac{\mu_0 I}{2\pi r} \implies \oint \underline{B} \cdot d\underline{l} = B 2\pi r = \frac{\mu_0 I}{2\pi r} \times 2\pi r = \mu_0 I$$

This gives us Ampere's Law:

$$\oint \underline{B} \cdot d\underline{l} = \mu_0 I_{\text{enclosed}}$$

2.1 Example

Consider a long solid cylinder carrying a uniformly distributed current. The cylinder has radius R and length

Thu 12 Mar 2026 11:00

Lecture 16 - Electromagnetism I: Induction

1 Ampere's Law Continued

2 Magnetic Inductance

Faraday suggested that, if currents induce magnetic fields, that the opposite might be possible. Could a magnetic field be used to induce a current? He found that whenever field lines move or change, an electric field is induced, which charges in the field respond to.

3 Motion of a Conductor in Magnetic Field

4 Induced E-Fields

5 Faraday's Law